Open Standards and Open Software for Geospatial Imagery

Open data, Cubesats, User Discovery, and Cloud-Optimized data formats

Philly Emerging Technologies for the Enterprise Conference 2020

Matthew Hanson **@GeoSkeptic**

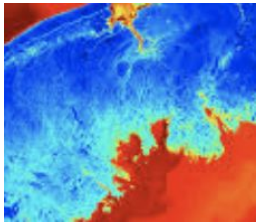
Element 84

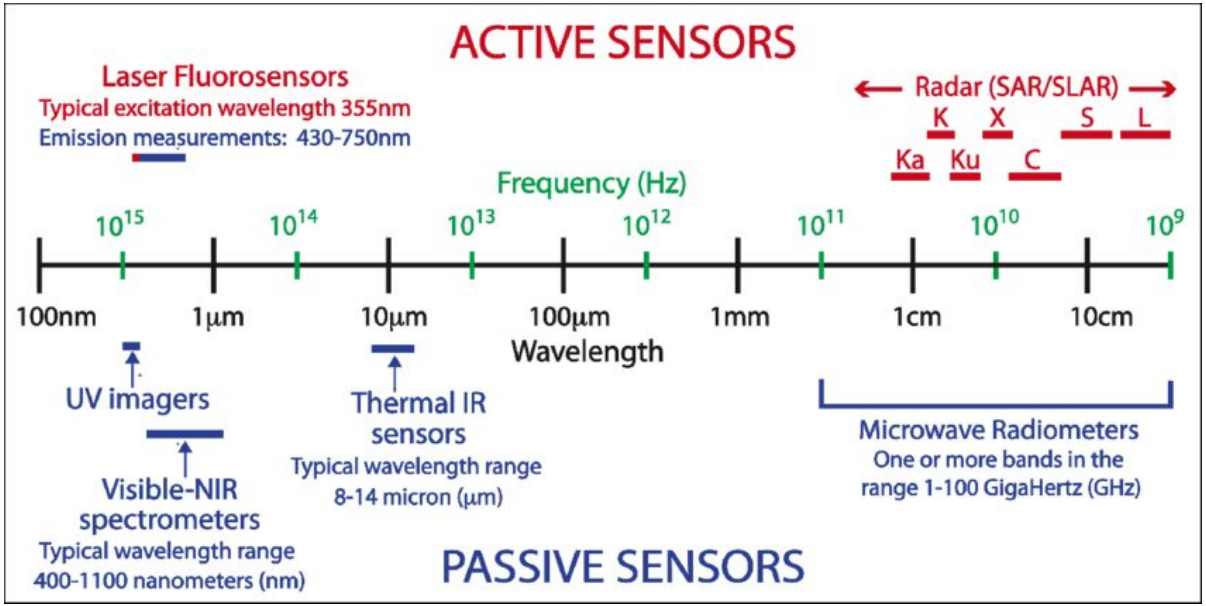
Geospatial Imagery

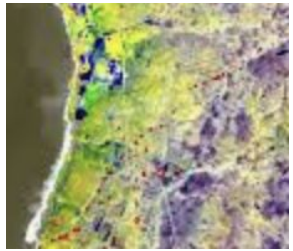
"raster data of the earth collected from a sensor"

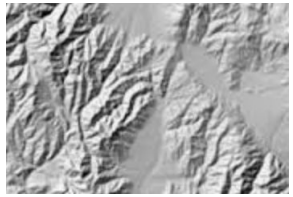


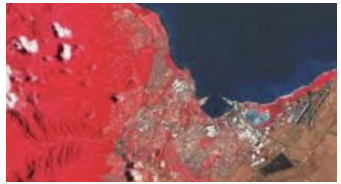












Remote Sensing

From wikipedia:

Remote sensing is the acquisition of information about an object or phenomenon without making physical contact with the object and thus in contrast to on-site observation, especially the Earth. Remote sensing is used in numerous fields, including geography, land surveying and most Earth science disciplines (for example, hydrology, ecology, meteorology, oceanography, glaciology, geology); it also has military, intelligence, commercial, economic, planning, and humanitarian applications.

Why Standards? Sounds boring

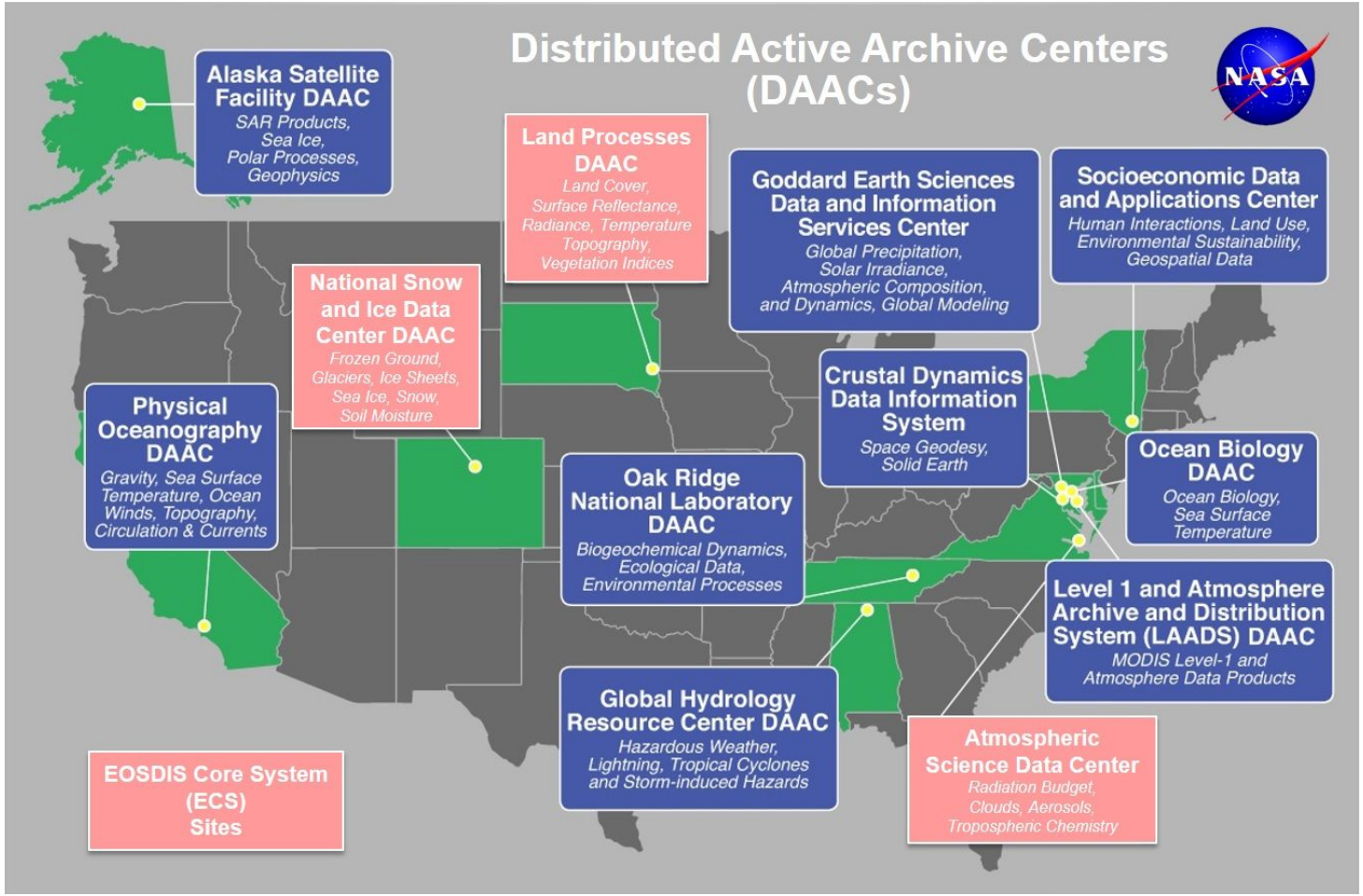
My background is in Remote Sensing

- Algorithm Development
- Analytics
- Machine learning
- Working closely with scientists

This has always been more difficult than it should be

- 80/20 rule: 80% is prepping data, 20% is interesting
- Need more and better tools …. But these need standards


NASA EARTH FLEET
OPERATING & FUTURE THROUGH 2023
TROPICS (6)
SENTINEL-6A/B
LANDSAT-9
NISAR
SWOT
TSIS-2
MAIA
PREFIRE (2)
TEMPO
GEOCARB
PACE
GLIMR
ICESAT-2
GRACE-FO (2)
CYGNSS (8)
NISTAR, EPIC (DSCOVR/NOAA)
SORCE
CLOUDSAT
TERRA
AQUA
AURA
CALIPSO
GPM
LANDSAT 7 (USGS)
LANDSAT 8 (USGS)
OCO-2
SMAP
SUOMI NPP (NOAA)
INVEST/CUBESATS
RAVAN
RainCube
CSIM
CubeRRT
TEMPEST-D
CIRiS
HARP
CTIM
HyTI
SNoOPI
NACHOS
ISS INSTRUMENTS
EMIT
CLARREO-PF
GEDI
SAGE III
OCO-3
TSIS-1
ECOSTRESS
LIS
JPSS-2, 3 & 4 INSTRUMENTS
OMPS-Limb
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(PRE) FORMULATION
IMPLEMENTATON
PRIMARY OPS
EXTENDED OPS

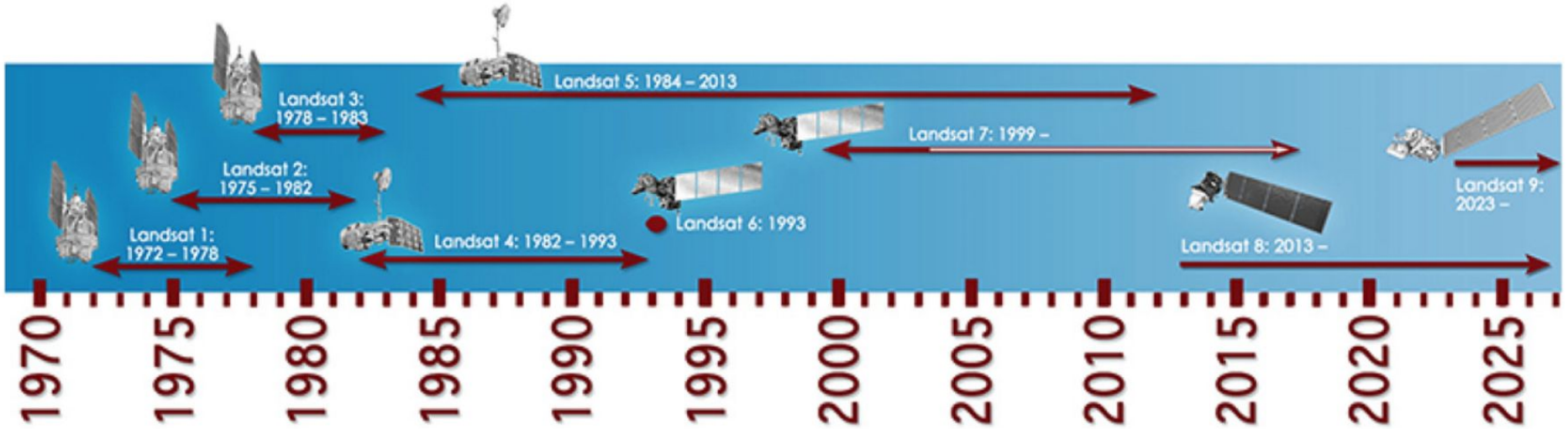

Distributed Active Archive Centers (DAACs)
NASA
Alaska Satellite Facility DAAC
SAR Products, Sea Ice, Polar Processes, Geophysics
Land Processes DAAC
Land Cover, Surface Reflectance, Radiance, Temperature Topography, Vegetation Indices
Goddard Earth Sciences Data and Information Services Center
Global Precipitation, Solar Irradiance, Atmospheric Composition, and Dynamics, Global Modeling
Socioeconomic Data and Applications Center
Human Interactions, Land Use, Environmental Sustainability, Geospatial Data
National Snow and Ice Data Center DAAC
Frozen Ground, Glaciers, Ice Sheets, Sea Ice, Snow, Soil Moisture
Physical Oceanography DAAC
Gravity, Sea Surface Temperature, Ocean Winds, Topography, Circulation & Currents
Crustal Dynamics Data Information System
Space Geodesy, Solid Earth
Ocean Biology DAAC
Ocean Biology, Sea Surface Temperature
Oak Ridge National Laboratory DAAC
Biogeochemical Dynamics, Ecological Data, Environmental Processes
Level 1 and Atmosphere Archive and Distribution System (LAADS) DAAC
MODIS Level-1 and Atmosphere Data Products
Global Hydrology Resource Center DAAC
Hazardous Weather, Lightning, Tropical Cyclones and Storm-induced Hazards
Atmospheric Science Data Center
Radiation Budget, Clouds, Aerosols, Tropospheric Chemistry
EOSDIS Core System (ECS) Sites

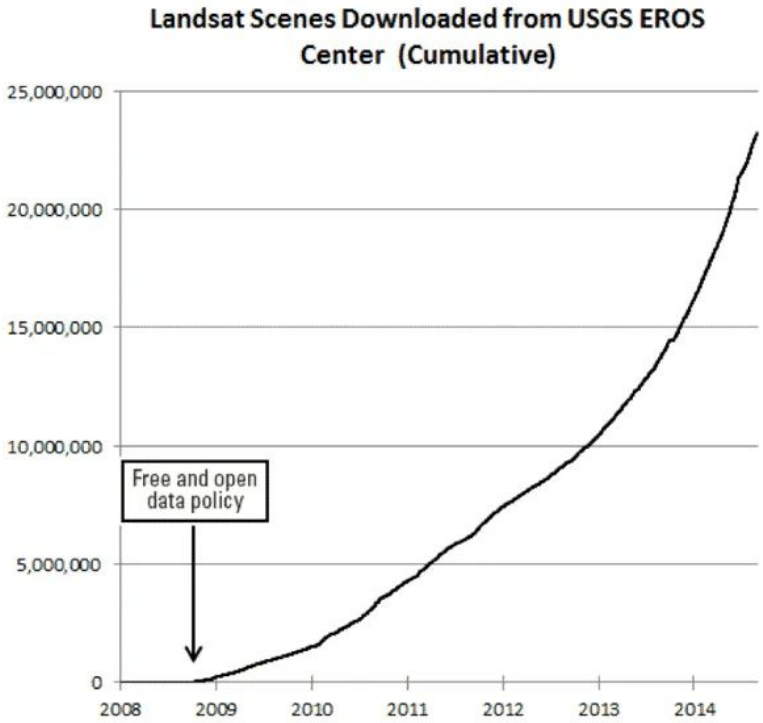
Free and Open Data Policies

NASA commits to the full and open sharing of Earth science data obtained from NASA Earth observing satellites, sub-orbital platforms and field campaigns with all users as soon as such data become available. There will be no period of exclusive access to NASA Earth science data.

Similar policies for Landsat data (USGS) and ESA data such as from the Sentinel satellites

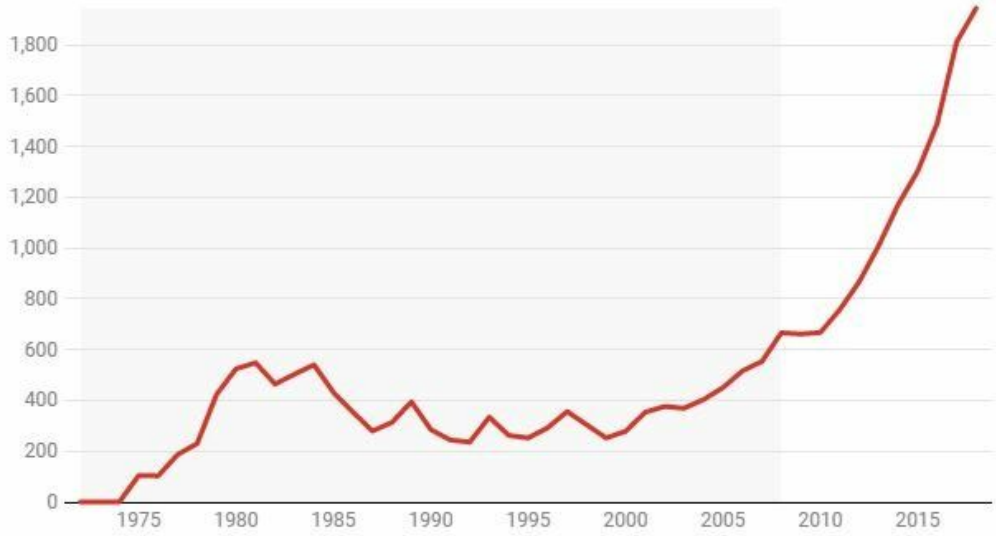
Landsat (USGS)





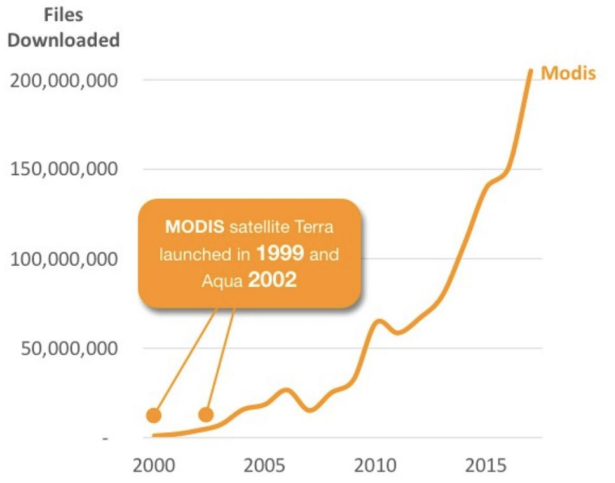
Landsat research

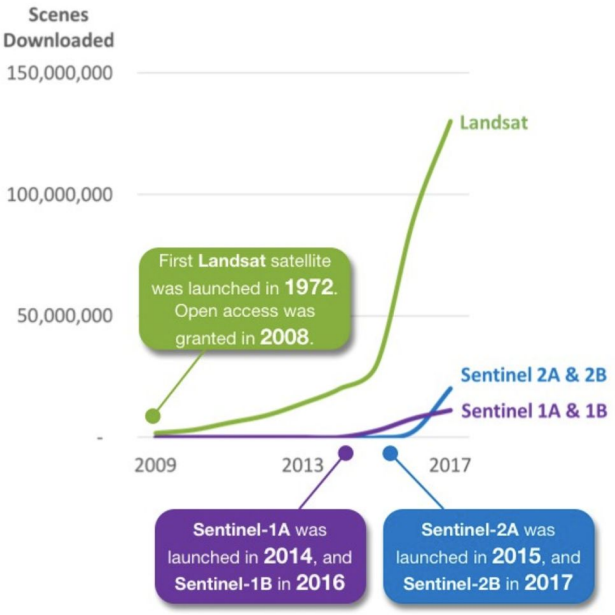
The number of publications per year citing Landsat imagery increased significantly after the U.S. Geological Survey made the data free in 2008.



Open Satellite Data Downloads

The number of downloads every year from the open data portals has increased exponentially!





SOURCES: MODIS and Landsat data was provided by NASA
Sentinel data was obtained via their annual reports, 2015-2017

*Files downloaded from the MODIS portal often contain multiple scenes

Radiant.Earth
Earth Imagery for Impact

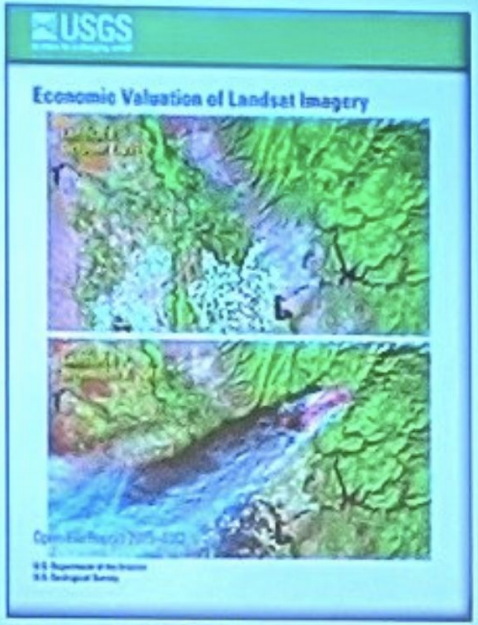
CC BY

Updated economic valuation of Landsat

"Landsat imagery provided domestic and international users an estimated **$3.45 billion** in benefits in 2017 compared to $2.19 billion in 2011, with U.S. users accounting for $2.06 billion of those benefits."

(Does not include value of scenes downloaded by cloud vendors or other downstream economic benefits such as value-added products)

USGS



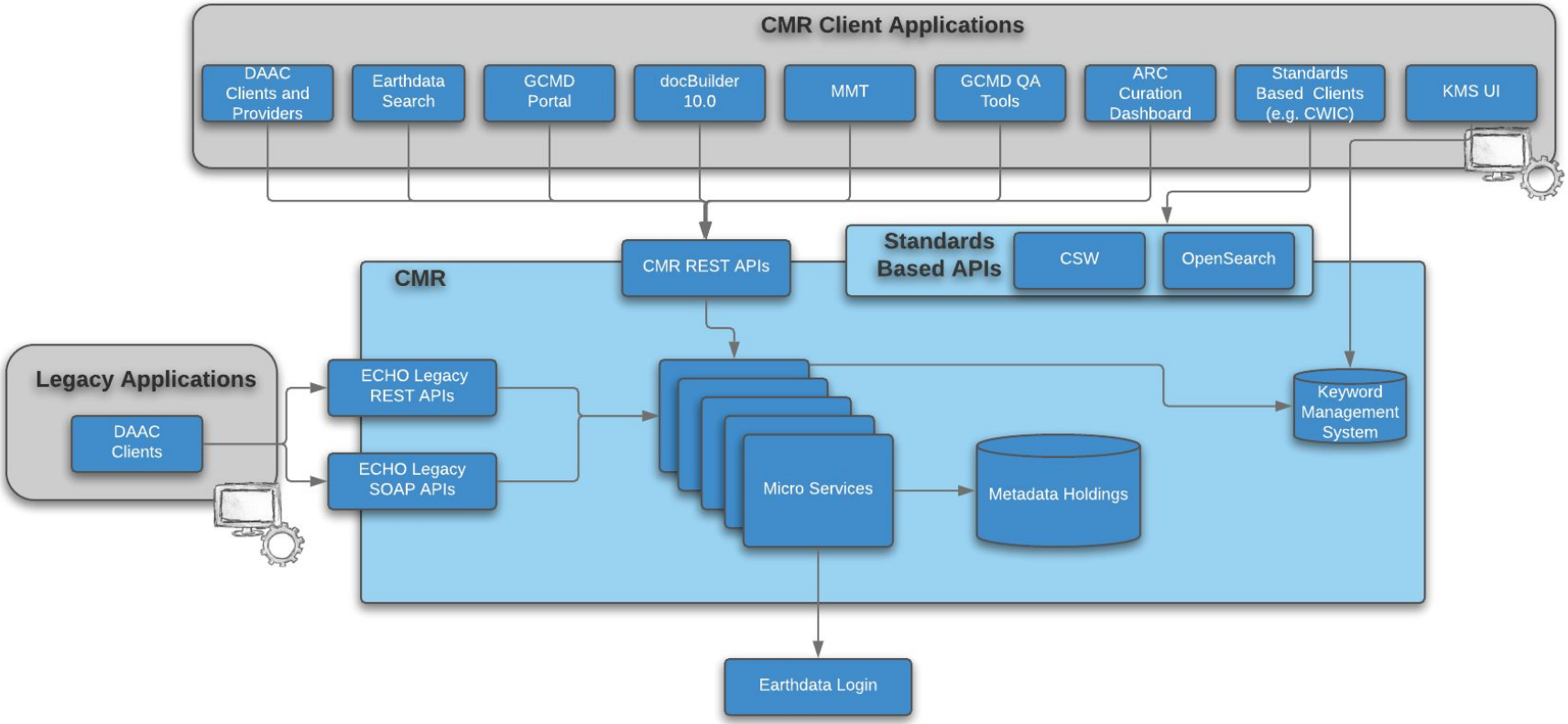
Economic Valuation of Landsat Imagery
Open-File Report 2019-1112
Crista L. Straub, Stephen R. Koontz, and John B. Loomis
https://doi.org/10.3133/ofr20191112


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TERRA
AQUA
AURA
CALIPSO
GPM
LANDSAT 7 (USGS)
LANDSAT 8 (USGS)
OCO-2
SMAP
SUOMI NPP (NOAA)
INVEST/CUBESATS
RAVAN
RainCube
CSIM
CubeRRT
TEMPEST-D
CIRiS
HARP
CTIM
HyTI
SNoOPI
NACHOS
ISS INSTRUMENTS
EMIT
CLARREO-PF
GEDI
SAGE III
OCO-3
TSIS-1
ECOSTRESS
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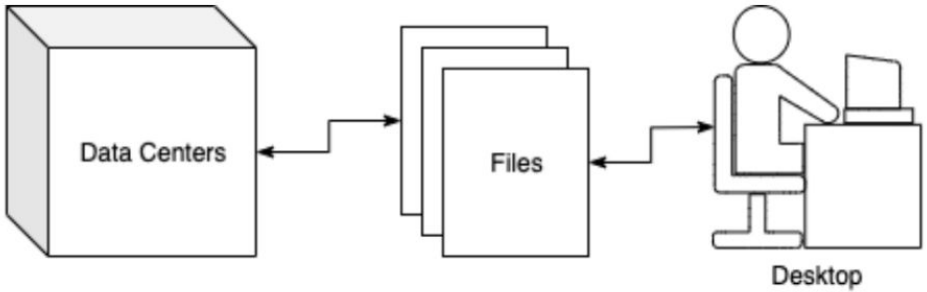
NASA Earth Science Data Distribution

- NASA Common Metadata Repository
  - All Earth Science Data from all DAACs
- Each DAAC also has own APIs and tools
- No single set of utilities for finding/fetching/using data
- Archival Data formats (not cloud-friendly)
  - Hierarchical Data Format (hdf)
  - NetCDF

NASA Common Metadata Repository

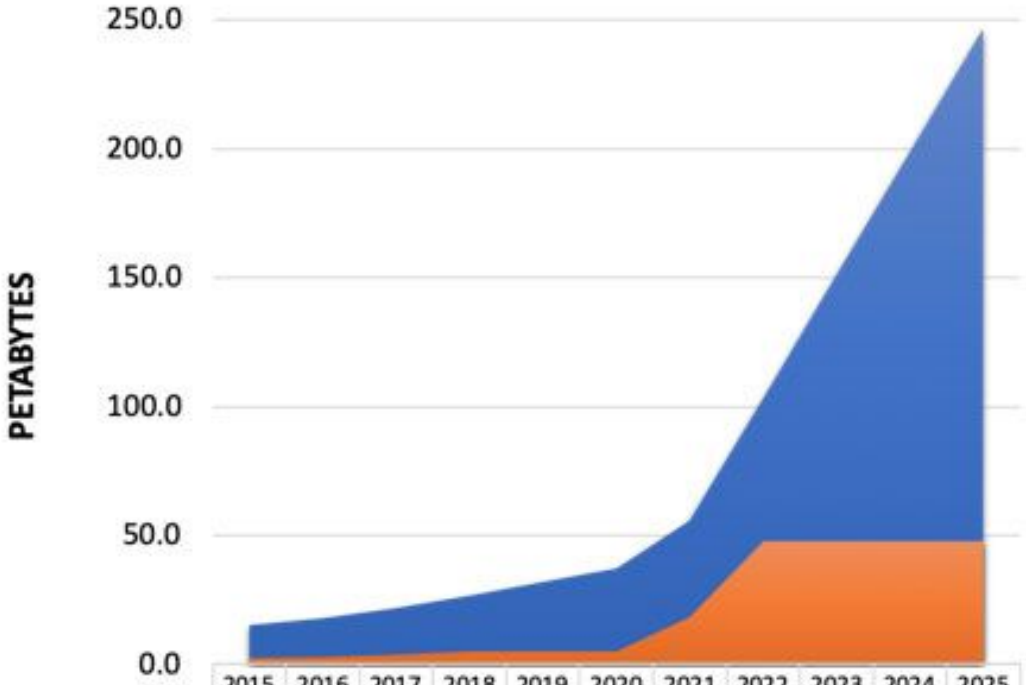


Traditional Workflow



- Search DAACs
- Download data files
- Process on desktop, or in HPC

NASA Earth Observation Archive



	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Archive Size (PB)	15.0	17.7	21.6	26.8	32.0	37.2	55.6	103.4	151.1	198.9	246.6
Annual Growth (PB)	2.6	2.8	3.9	5.2	5.2	5.2	18.4	47.7	47.7	47.7	47.7

Challenges

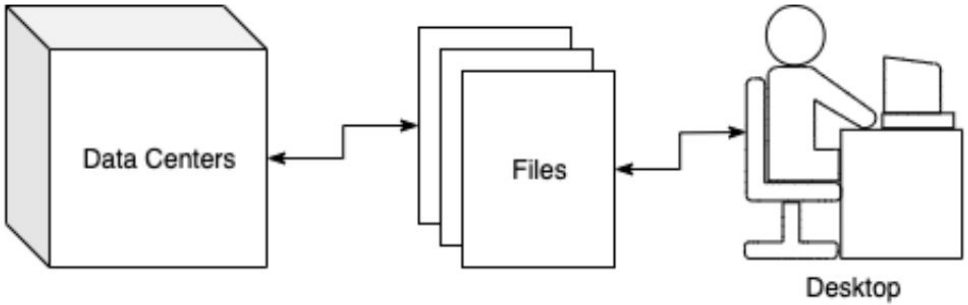
There is a wealth of Free and Open data available from NASA, ESA, USGS

Yet...

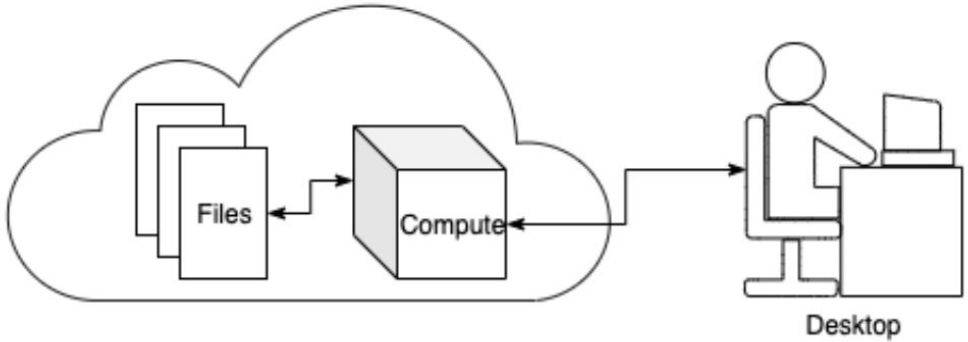
finding the right data
fetching it
pre-processing it

is difficult to scale due to data volume and dataset differences

Changing the Workflow

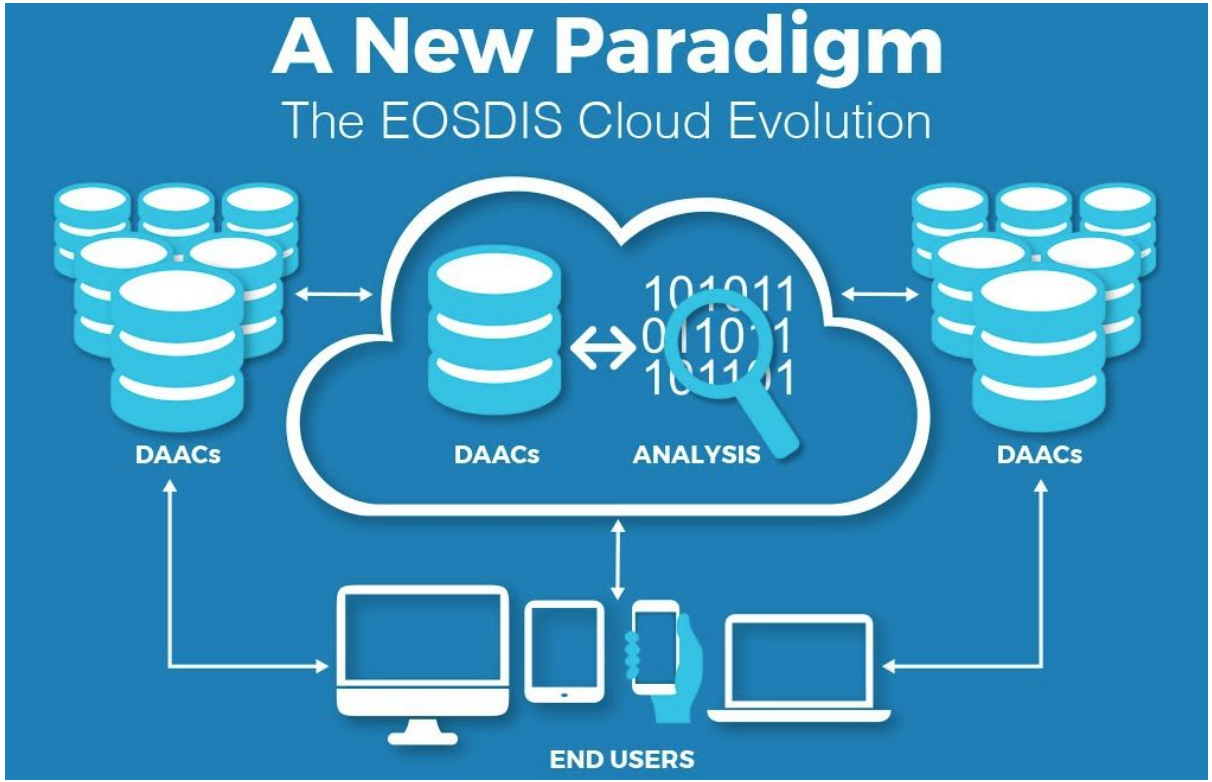


Local Processing



Cloud Processing

NASA Cumulus: Moving to the Cloud

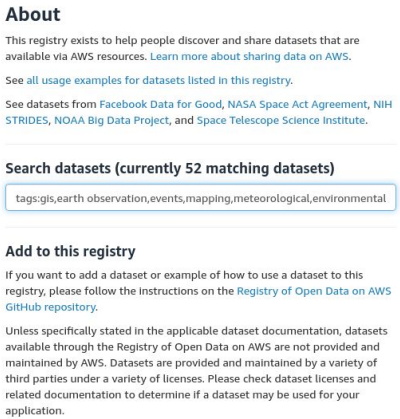


Earth on AWS

Copies of high value NASA/USGS/ESA datasets

Cloud-optimized data formats

Process data on instances in same-region as data



About

This registry exists to help people discover and share datasets that are available via AWS resources. Learn more about sharing data on AWS.

See all usage examples for datasets listed in this registry.

See datasets from Facebook Data for Good, NASA Space Act Agreement, NIH STRIDES, NOAA Big Data Project, and Space Telescope Science Institute.

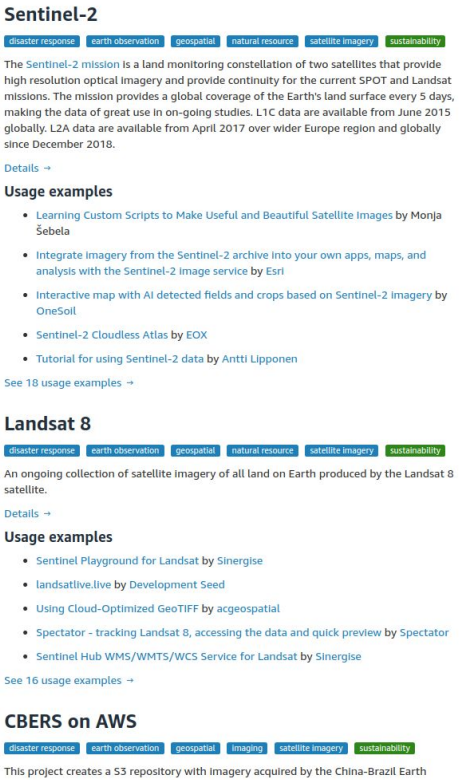
Search datasets (currently 52 matching datasets)

tags:gis,earth observation,events,mapping,meteorological,environmental

Add to this registry

If you want to add a dataset or example of how to use a dataset to this registry, please follow the instructions on the Registry of Open Data on AWS GitHub repository.

Unless specifically stated in the applicable dataset documentation, datasets available through the Registry of Open Data on AWS are not provided and maintained by AWS. Datasets are provided and maintained by a variety of third parties under a variety of licenses. Please check dataset licenses and related documentation to determine if a dataset may be used for your application.



Sentinel-2

disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

The Sentinel-2 mission is a land monitoring constellation of two satellites that provide high resolution optical imagery and provide continuity for the current SPOT and Landsat missions. The mission provides a global coverage of the Earth's land surface every 5 days, making the data of great use in on-going studies. L1C data are available from June 2015 globally. L2A data are available from April 2017 over wider Europe region and globally since December 2018.

Details →

Usage examples

- Learning Custom Scripts to Make Useful and Beautiful Satellite Images by Monja Šebela
- Integrate imagery from the Sentinel-2 archive into your own apps, maps, and analysis with the Sentinel-2 image service by Esri
- Interactive map with AI detected fields and crops based on Sentinel-2 imagery by OneSoil
- Sentinel-2 Cloudless Atlas by EOX
- Tutorial for using Sentinel-2 data by Antti Lipponen

See 18 usage examples →

Landsat 8

disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

An ongoing collection of satellite imagery of all land on Earth produced by the Landsat 8 satellite.

Details →

Usage examples

- Sentinel Playground for Landsat by Sinergise
- landsatlive.live by Development Seed
- Using Cloud-Optimized GeoTIFF by acgeospatial
- Spectator - tracking Landsat 8, accessing the data and quick preview by Spectator
- Sentinel Hub WMS/WMTS/WCS Service for Landsat by Sinergise

See 16 usage examples →

CBERS on AWS

disaster response | earth observation | geospatial | imaging | satellite imagery | sustainability

This project creates a S3 repository with imagery acquired by the China-Brazil Earth

Commercial Remote Sensing Data

- Higher resolution
- Taskable
- Also...Expensive

In 2013, Planet (Labs) launched 2 small satellites

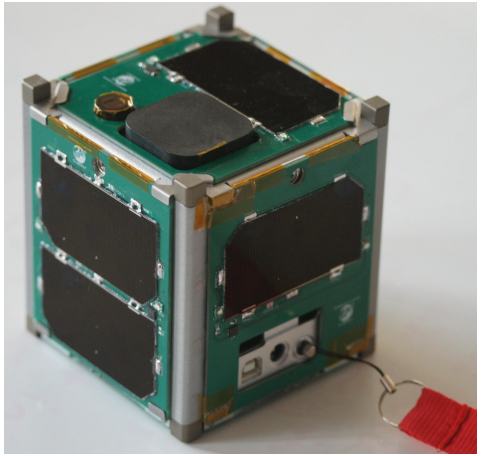
This kicked off the CubeSat era

What are CubeSats?

Satellite mass classification

- Large satellites: >1000 kg
- Medium satellites: 500 to 1000 kg
- Small satellites: < 500 kg
  - Minisatellites: 100 to 500 kg
  - Microsatellites: 10 to 100 kg
  - Nanosatellites: 1 to 10 kg
  - Picosatellites: 100 g – 1 kg
  - Femtosatellites: 10 g – 100 g
  - Attosatellites: 1 g – 10 g
  - Zeptosatellites: 0.1 g – 1 g

- CubeSat sizes:
  - From ~0.2 kg to ~40 kg
  - From 0.25U to 27U

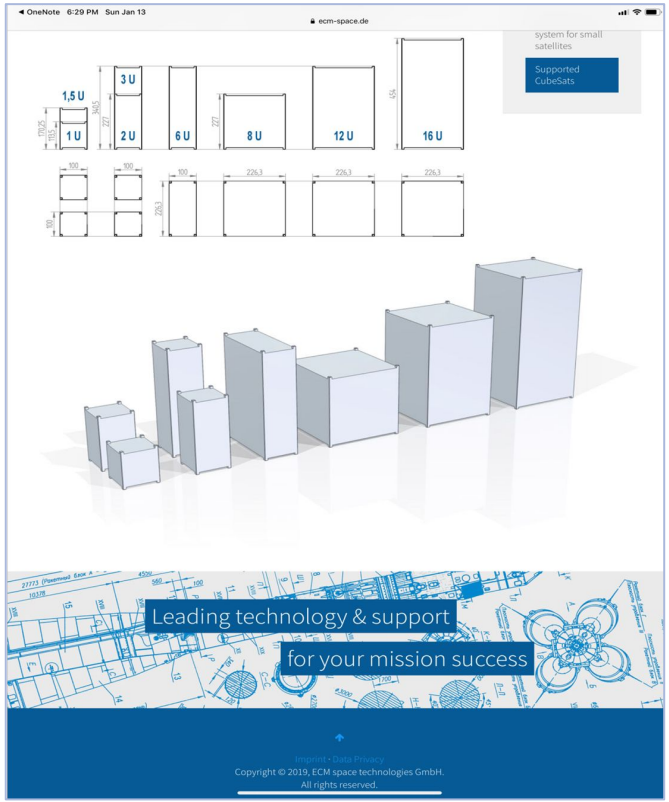


Strict specifications allows for mass production of parts, reducing cost.

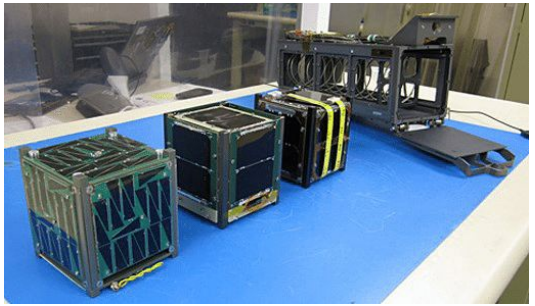
By Cbrandonvt - Own work, CC BY 3.0
https://commons.wikimedia.org/w/index.php?curid=37677790

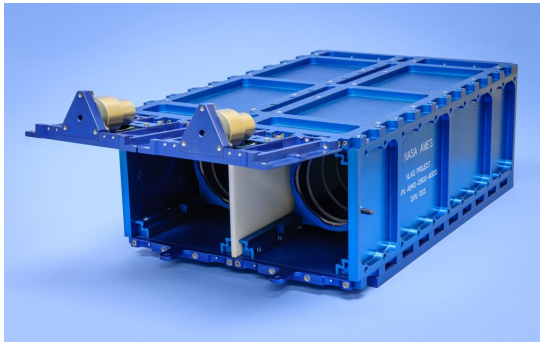
nanosats.eu/cubesat.html

CubeSat Dispenser Systems

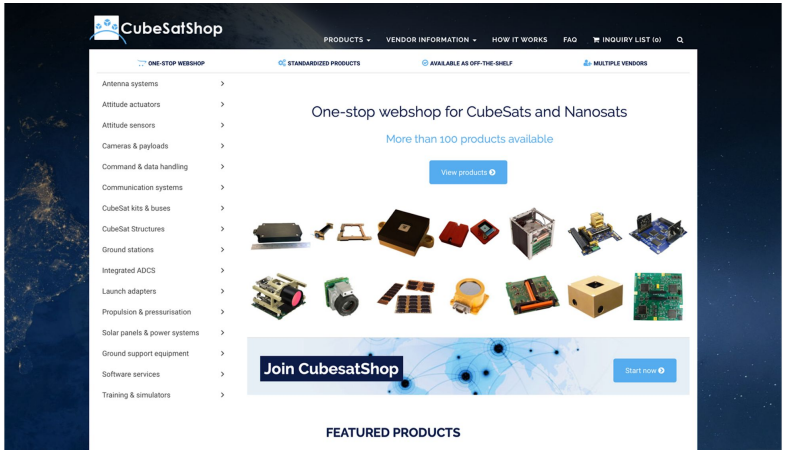


- Poly-Picosatellite Orbital Deployer (P-POD) supports up to 3U of CubeSats.
- Bolted onto whatever won't fall off.
- Launch vehicle says when to release, door opens, satellites stream out.

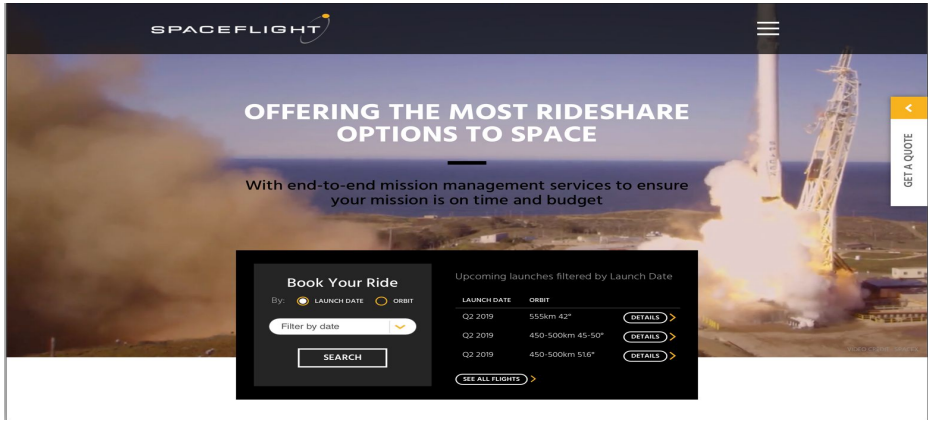




So you want a CubeSat ?



cubesatshop.com

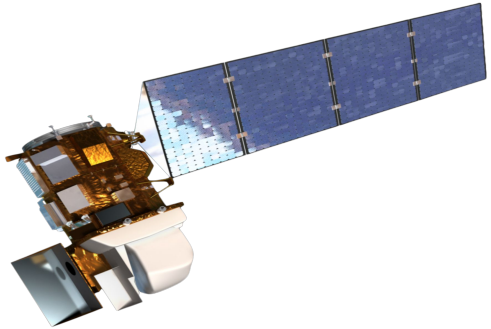


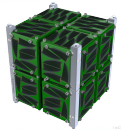
spaceflight.com

Commercial-Off-The-Shelf is a legitimate option for building and launching CubeSats

Price Tag: $350,000 +

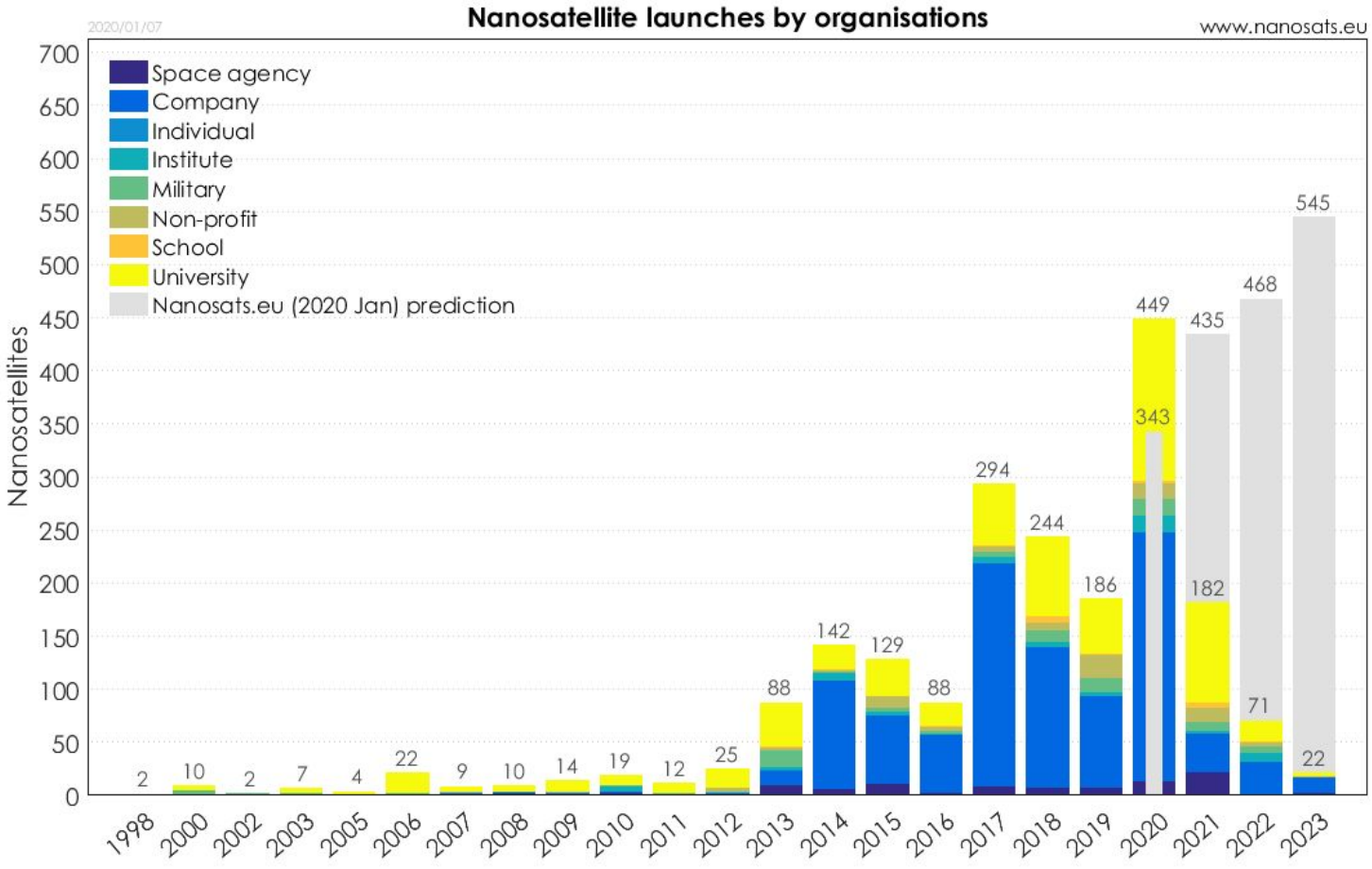
Traditional Sats vs CubeSats





- Baseline science quality data
- 2623kg
- 3m long, 2.4m diameter
- ~$850 billion
- 5+ year design life
  (Landsat 5 flew for 29 years)

- Experiments, rapid iteration, “good enough” data
- ~1-1.33kg
- 10cm Cube
- ~$40k + Launch Fees
- Weeks to months design life


Nanosatellite launches by organisations
2020/01/07
www.nanosats.eu
Nanosatellites
Space agency
Company
Individual
Institute
Military
Non-profit
School
University
Nanosats.eu (2020 Jan) prediction
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2023

Now what ?

Thanks to standards, we have a CubeSat

What do we do with data ?

- Search and discovery?
- Analytics?

Powered By
Element 84

CubeSatData.com is a hosted service for managing cube and small satellite data.

CubeSat Data can manage the entire data pipeline of your cube or small satellite mission -- from ground station downlink and data storage to automated archives of convenience so you can put your data to work solving problems.

LET'S TALK CUBESAT!



FilmDrop

Satellite Data Pipeline Processing and Management

Data is stored and indexed with mission-optimized, open data and metadata standards.



Search, discover, and visualize your data at scale and make it available via services for your team or customers.

Only write processing functions, FilmDrop deploys and scales

Element 84

As with launching CubeSats, the answer is

STANDARDS

- Satellite downlink
- Processing
- Metadata
- APIs
- Data Formats

Open Geospatial Consortium (OGC) Standards

OGC API (New)	OGC Legacy (Old)	What is it for?
Common	OWS Common	Common API elements
Features	Web Feature Service (WFS)	Serving geospatial vector data
Coverages	Web Coverage Services (WCS)	Serving geospatial raster data
Maps	Web Map Service (WMS)	Serving rendered map layers
Tiles	Web Map Tile Service (WMTS)	Serving data as tiles
Processes	Web Processing Service (WPS)	Invoking geospatial processing requests
Records	Catalog Service for the Web (CSW)	Searching metadata

OGC APIs

- Ongoing work
- OGC API - Features 1.0 released

See **ogcapi.org**

Technological Gaps (circa 2018)

If your [meta]data cannot be crawled and indexed, your data might as well not exist

If you need to download files, you will not be able to scale processing

If data provenance cannot be determined, your research cannot be reproduced

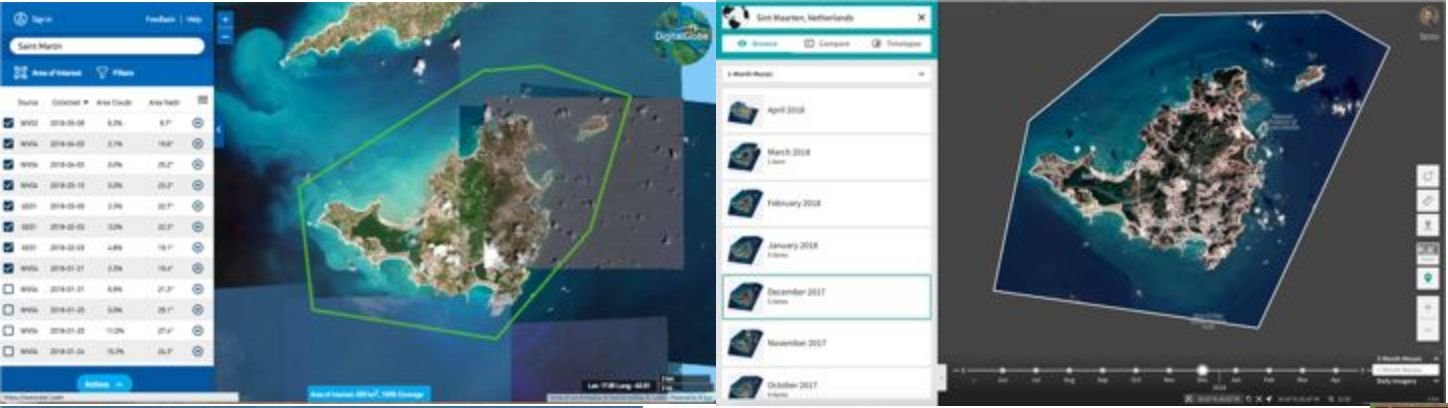
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LC08_L1GT_168005_20190828_20190828_01_RT.tif
LC08_L1GT_168005_20190828_20190828_01_RT_B1.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_B3.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_c10_f1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_cm.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_class_c50_i9.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_tmp_test5.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_ver7.6.tif
LC08_L1GT_168005_20190828_20190828_01_final_class2.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_class.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_final.tif
LC08_L1GT_168005_20190828_20190828_01_test.tif
LC08_L1GT_168005_20190828_20190828_test1.tif
LC08_L1GT_168005_20190828_20190828_test_test.tif
```

SpatioTemporal Asset Catalogs

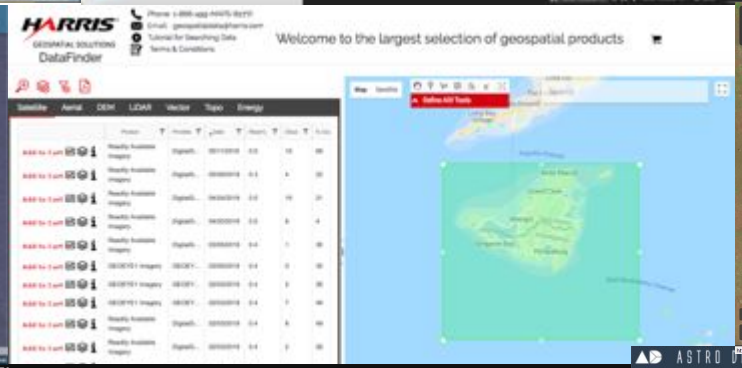


A specification to improve interoperability of geospatial data

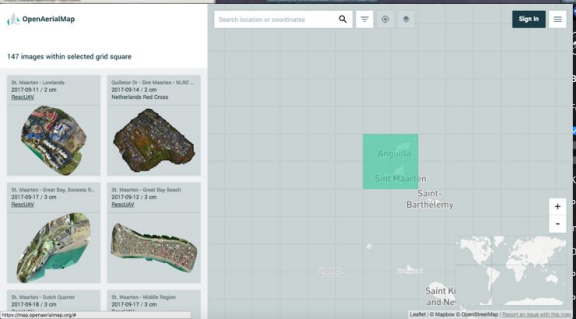
Why?


Saint Martin
Sint Maarten, Netherlands
April 2018
March 2018
February 2018
January 2018
December 2017
November 2017
October 2017

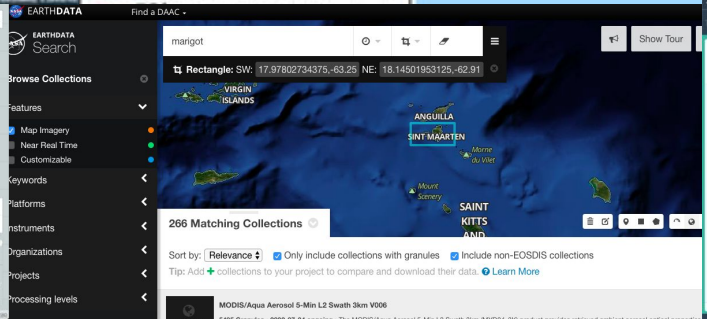

SENTINEL Hub Playground

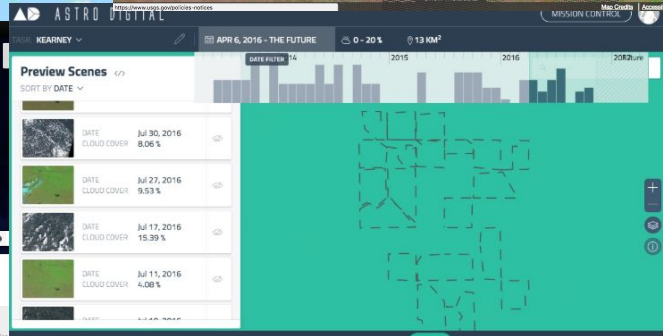

HARRIS GEOSPATIAL SOLUTIONS
DataFinder
Welcome to the largest selection of geospatial products

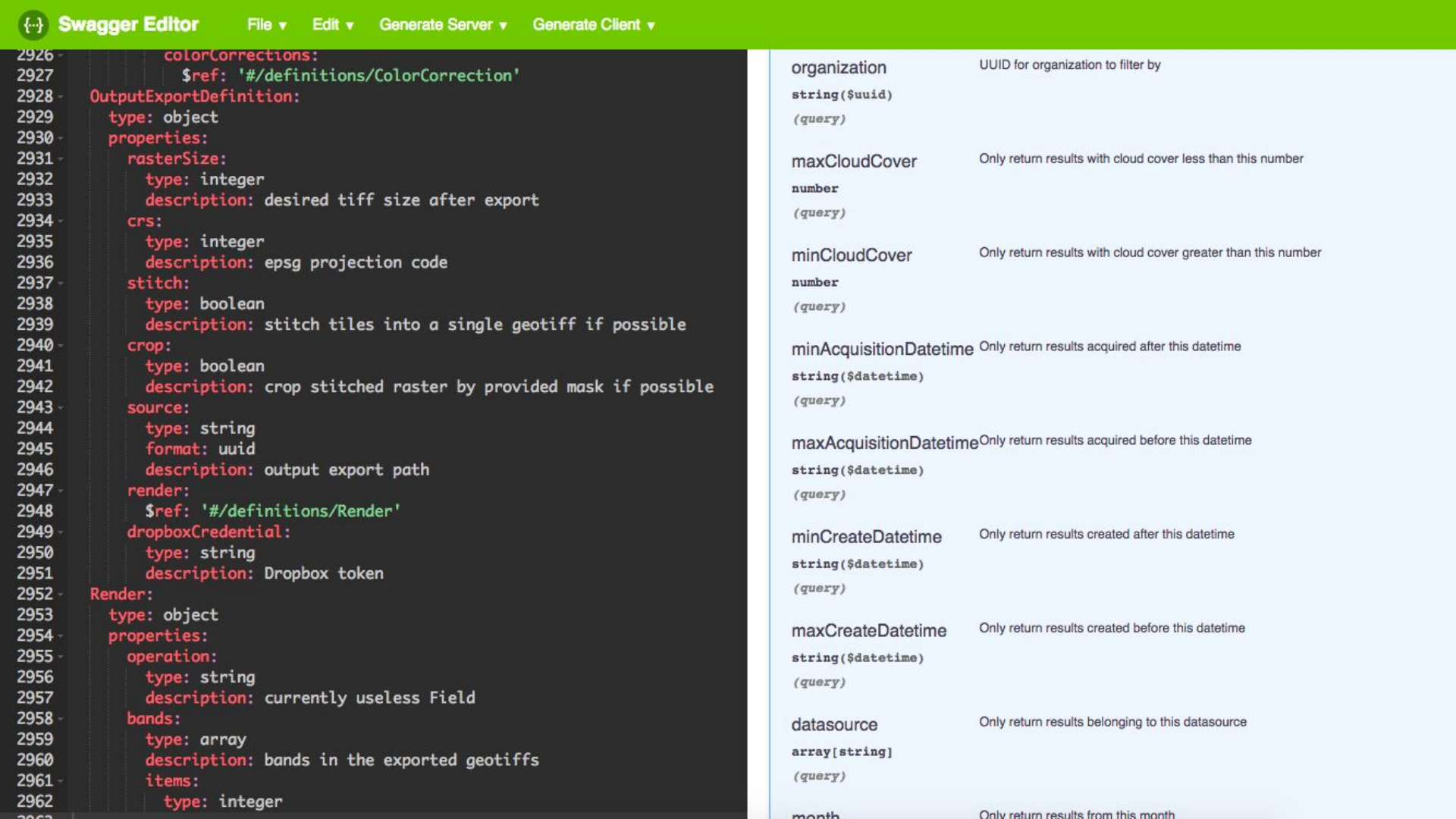

USGS LandsatLook
Show Metadata


OpenAerialMap
147 images within selected grid square

Anguilla
Sint Maarten
Saint-Barthelemy

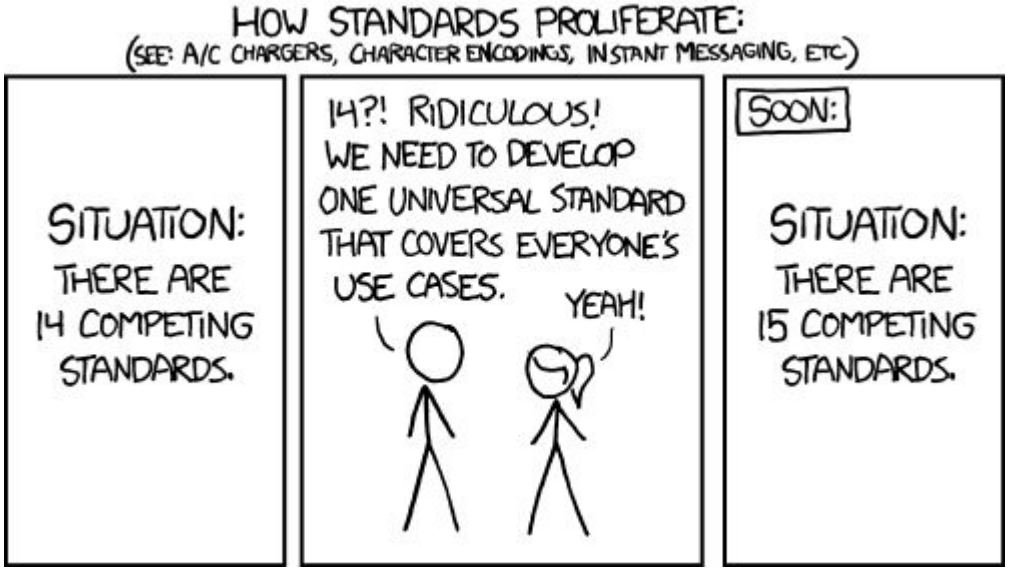

EARTHDATA
Search
marigot
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91
Show Tour
Browse Collections
Features
Map Imagery
Near Real Time
Customizable
Keywords
Platforms
Instruments
Organizations
Projects
Processing levels
VIRGIN ISLANDS
ANGUILLA
SINT MAARTEN
SAINT KITTS AND
266 Matching Collections
Sort by: Relevance
Only include collections with granules
Include non-EOSDIS collections
MODIS/Aqua Aerosol 5-Min L2 Swath 3km V006


ASTRO DIGITAL
MISSION CONTROL
KEARNEY
APR 6, 2016 - THE FUTURE
0 - 20 %
13 KM²
Preview Scenes
SORT BY DATE
Jul 30, 2016
8.06 %
Jul 27, 2016
9.93 %
Jul 17, 2016
15.39 %
Jul 11, 2016
4.08 %


Swagger Editor
File
Edit
Generate Server
Generate Client
colorCorrections:
$ref: '#/definitions/ColorCorrection'
OutputExportDefinition:
type: object
properties:
rasterSize:
type: integer
description: desired tiff size after export
crs:
type: integer
description: epsg projection code
stitch:
type: boolean
description: stitch tiles into a single geotiff if possible
crop:
type: boolean
description: crop stitched raster by provided mask if possible
source:
type: string
format: uuid
description: output export path
render:
$ref: '#/definitions/Render'
dropboxCredential:
type: string
description: Dropbox token
Render:
type: object
properties:
operation:
type: string
description: currently useless Field
bands:
type: array
description: bands in the exported geotiffs
items:
type: integer
organization
UUID for organization to filter by
string($uuid)
(query)
maxCloudCover
Only return results with cloud cover less than this number
number
(query)
minCloudCover
Only return results with cloud cover greater than this number
number
(query)
minAcquisitionDatetime
Only return results acquired after this datetime
string($datetime)
(query)
maxAcquisitionDatetime
Only return results acquired before this datetime
string($datetime)
(query)
minCreateDatetime
Only return results created after this datetime
string($datetime)
(query)
maxCreateDatetime
Only return results created before this datetime
string($datetime)
(query)
datasource
Only return results belonging to this datasource
array[string]
(query)

STAC

SpatioTemporal Asset Catalog



How to avoid this - build a community around the standard



Who?

Element 84

Radiant Earth Foundation
EARTH IMAGERY FOR IMPACT

azavea

MAXAR TECHNOLOGIES

Element 84

open EO

developmentSEED

THE CLIMATE CORPORATION

HARRIS

Australian Government
Geoscience Australia

planet.

Google Earth Engine

CHINA-BRAZIL EARTH RESOURCES SATELLITE
CBERS
INPE CAST

SpaceNet

astræa
See the Earth as it could be.

HEXAGON GEOSPATIAL

geo

Boundless

stacspec.org

What is STAC?

SpatioTemporal Asset Catalog

- Defined metadata for geospatial catalogs and assets
  - Focus on search and discovery
  - Simple and extensible
- Linked metadata
  - Crawlable so can be indexed
  - Maintain data provenance
- Dynamic API
  - Searchable on arbitrary fields
  - OGC API - Features with STAC specific extensions

What is it not?

SpatioTemporal Asset Catalog

- Full-fledged metadata standard
- Replacement for data provider's internal metadata
- Single source of truth
- Replacement for ISO 19115, OGC CSW, …

The specification

- Catalogs
  - There's always a root catalog
  - Contains other catalogs and collections
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Ultimately contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when and where)

Links

Field Name	Type	Description
href	string	**REQUIRED.** The actual link in the format of an URL. Relative and absolute links are both allowed.
rel	string	**REQUIRED.** Relationship between the current document and the linked document. See chapter "Relation types" for more information.
type	string	Media type of the referenced entity.
title	string	A human readable title to be used in rendered displays of the link.

Link Relation Types

Type	Description
self	STRONGLY RECOMMENDED. *Absolute* URL to the location that the catalog file can be found online, if available. This is particularly useful when in a download package that includes metadata, so that the downstream user can know where the data has come from.
root	STRONGLY RECOMMENDED. URL to the root STAC Catalog or Collection. Catalogs should include a link to their root, even if it's the root and points to itself.
parent	URL to the parent STAC Catalog or Collection. Non-root catalogs should include a link to their parent.
child	URL to a child STAC Catalog or Collection.
item	URL to a STAC Item.

Other Links: **about, derived-from, previous-version (any IANA rel type)**

Item Fields

Field Name	Type	Description
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. As most geospatial assets are already defined by some identification scheme by the data provider it is recommended to simply use that ID. Data providers are advised to include sufficient information to make their IDs globally unique, including things like unique satellite IDs.
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
geometry	GeoJSON Geometry Object	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED.** Bounding Box of the asset represented by this item, formatted according to RFC 7946, section 5.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is strongly recommended.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present. This field provides an easy way for a user to search for any Items that belong in a specified Collection.

Item Properties

Field Name	Type	Description
datetime	string	**REQUIRED.** The searchable date and time of the assets, in UTC. It is formatted according to RFC 3339, section 5.6.
title	string	A human readable title describing the item.
start_datetime	string	The first or start date and time for the item, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.
end_datetime	string	The last or end date and time for the item, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.

Field Name	Type	Description
platform	string	Unique name of the specific platform to which the instrument is attached.
instruments	[string]	Name of instrument or sensor used (e.g., MODIS, ASTER, OLI, Canon F-1).
constellation	string	Name of the constellation to which the platform belongs.
mission	string	Name of the mission for which data is collected.

Item Assets

Field Name	Type	Description
href	string	**REQUIRED.** Link to the asset object. Relative and absolute links are both allowed.
title	string	The displayed title for clients and users.
description	string	A description of the Asset providing additional details, such as how it was processed or created.
type	string	Media type of the asset.
roles	[string]	The semantic role of the asset, similar to the use of `rel` in links.

Content Extensions

Extension Name (Prefix)	Scope	Description	Maturity
Asset Definition (-)	Collection	Provides a way to specify details about what assets may be found in Items belonging to a collection.	*Proposal*
Checksum (`checksum`)	Item, Catalog, Collection	Provides a way to specify file checksums for assets and links in Items, Catalogs and Collections.	*Proposal*
Commons (-)	Item, Collection	Provides a way to specify data fields in a collection that are common across the STAC Items in that collection, so that each does not need to repeat all the same information.	*Proposal*
Data Cube (`cube`)	Item, Collection	Data Cube related metadata, especially to describe their dimensions.	*Proposal*
EO (`eo`)	Item	Covers electro-optical data that represents a snapshot of the earth for a single date and time. It could consist of multiple spectral bands, for example visible bands, infrared bands, red edge bands and panchromatic bands. The extension provides common fields like bands, cloud cover, gsd and more.	*Pilot*
Label (`label`)	Item	Items that relate labeled AOIs with source imagery	*Proposal*
Point Cloud (`pc`)	Item	Provides a way to describe point cloud datasets. The point clouds can come from either active or passive sensors, and data is frequently acquired using tools such as LiDAR or coincidence-matched imagery.	*Proposal*
Projection (`proj`)	Item	Provides a way to describe items whose assets are in a geospatial projection.	*Proposal*
SAR (`sar`)	Item	Covers synthetic-aperture radar data that represents a snapshot of the earth for a single date and time.	*Proposal*
Satellite (`sat`)	Item	Satellite related metadata for data collected from satellites.	*Proposal*
Scientific (`sci`)	Item, Collection	Scientific metadata is considered to be data that indicate from which publication data originates and how the data itself should be cited or referenced.	*Proposal*
Single File STAC (-)	ItemCollection	An extension to provide a set of Collections and Items as a single file catalog.	*Proposal*
Versioning Indicators (-)	Item, Collection	Provides fields and link relation types to provide a version and indicate deprecation.	*Proposal*

EO Extension

Field Name	Type	Description
eo:gsd	number	**REQUIRED.** Ground Sample Distance at the sensor.
eo:bands	[Band Object]	**REQUIRED.** This is a list of the available bands where each item is a Band Object.
eo:cloud_cover	number	Estimate of cloud cover as a percentage (0-100) of the entire scene. If not available the field should not be provided.

EO Band Object

Field Name	Type	Description
name	string	The name of the band (e.g., "B01", "B02", "B1", "B5", "QA").
common_name	string	The name commonly used to refer to the band to make it easier to search for bands across instruments. See the list of accepted common names.
description	string	Description to fully explain the band. CommonMark 0.29 syntax MAY be used for rich text representation.
center_wavelength	number	The center wavelength of the band, in micrometers (µm).
full_width_half_max	number	Full width at half maximum (FWHM). The width of the band, as measured at half the maximum transmission, in micrometers (µm).

EO Band Common Names

Common Name	Band Range (µm)	Landsat 5/7	Landsat 8	Sentinel 2	MODIS
coastal	0.40 - 0.45		1	1	
blue	0.45 - 0.50	1	2	2	3
green	0.50 - 0.60	2	3	3	4
red	0.60 - 0.70	3	4	4	1
yellow	0.58 - 0.62				
pan	0.50 - 0.70	8 (*L7 only*)	8		
rededge	0.70 - 0.75				
nir	0.75 - 1.00	4		8	2
nir08	0.75 - 0.90		5	8a	
nir09	0.85 - 1.05			9	
cirrus	1.35 - 1.40		9	10	26
swir16	1.55 - 1.75	5	6	11	6
swir22	2.10 - 2.30	7	7	12	7
lwir	10.5 - 12.5	6			
lwir11	10.5 - 11.5		10		31
lwir12	11.5 - 12.5		11		32

Linking Assets to Bands

```
- eo:bands: [
    - {
        name: "B1",
        common_name: "coastal",
        gsd: 30,
        center_wavelength: 0.44,
        full_width_half_max: 0.02
      },
    - {
        name: "B2",
        common_name: "blue",
        gsd: 30,
        center_wavelength: 0.48,
        full_width_half_max: 0.06
      },
    - {
        name: "B3",
        common_name: "green",
        gsd: 30,
        center_wavelength: 0.56,
        full_width_half_max: 0.06
      },
```

```
- assets: {
    - index: {
        type: "text/html",
        title: "HTML index page"
      },
    - thumbnail: {
        title: "Thumbnail image",
        type: "image/jpeg"
      },
    - B1: {
        type: "image/x.geotiff",
      - eo:bands: [
            0
        ],
        title: "Band 1 (coastal)"
      },
    - B2: {
        type: "image/x.geotiff",
      - eo:bands: [
            1
        ],
        title: "Band 2 (blue)"
      },
    - B3: {
        type: "image/x.geotiff",
      - eo:bands: [
            2
        ],
        title: "Band 3 (green)"
      },
```

https://earth-search.aws.element84.com/stac

+ - View source

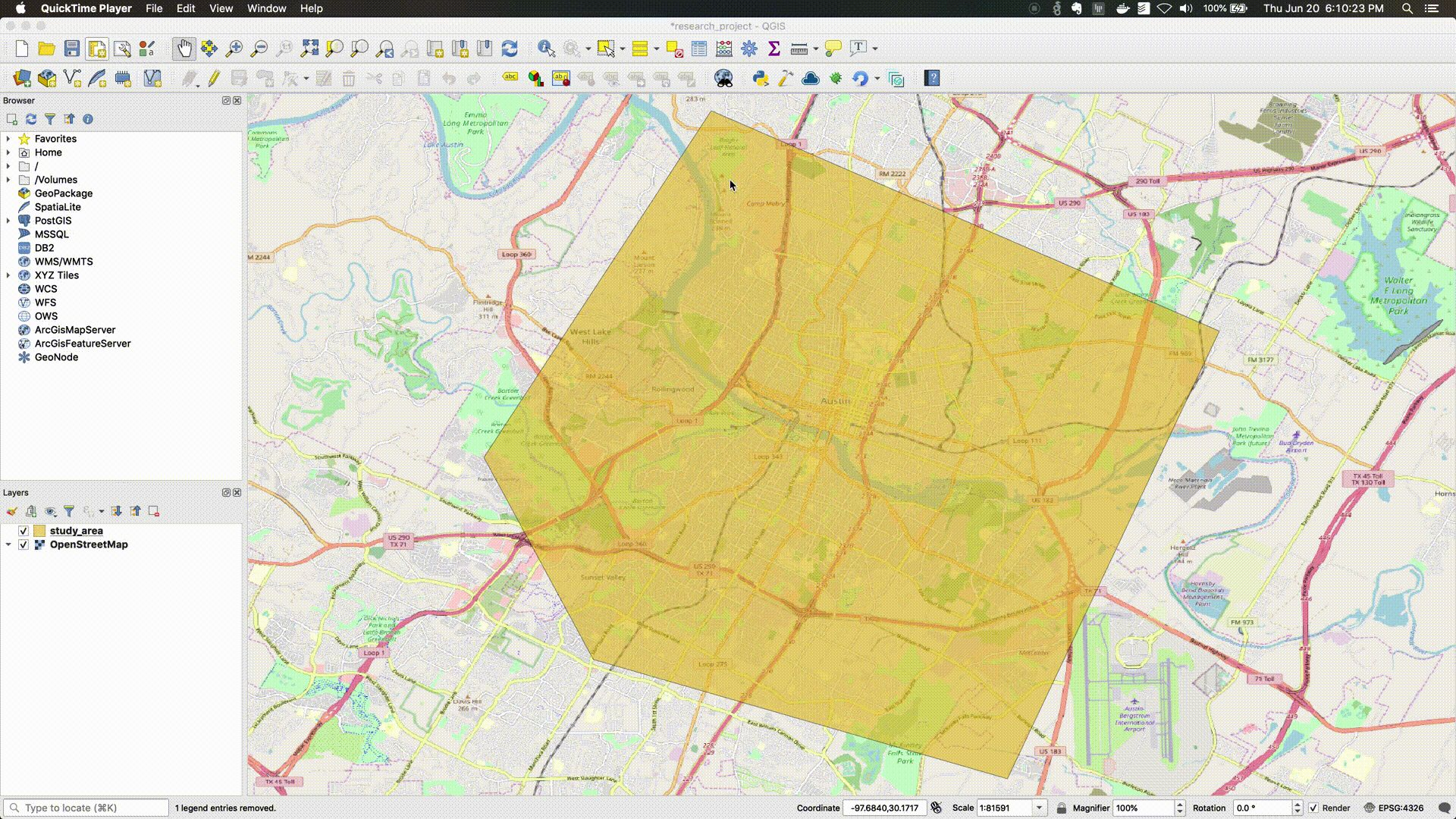

```
{
   stac_version: "0.6.2",
   id: "earth-search",
   title: "Earth on AWS STAC API",
   description: "A STAC API powered by sat-api for Earth on AWS datasets by Element 84",
 - links: [
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-awfi"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan10m"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-mux"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/sentinel-2-l1c"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/landsat-8-l1"
       },
     - {
           rel: "child",
           href: "https://earth-search.aws.element84.com/collections/cbers4-pan5m"
       },
     - {
           rel: "self",
           href: "https://earth-search.aws.element84.com/stac"
       }
   ]
}
```


links

STAC Open-Source Ecosystem

Implementation-driven development

- Validators and readers: stac-lint, PySTAC, sat-stac
- APIs: stac-api, staccato, franklin
- Browsers: stac-browser, sat-api-browser, QGIS Plugin
- Clients: sat-search, sat-fetch
- Catalogs: NASA-CMR STAC, E84 Earth-search, SpaceNet, CBERS-4


QuickTime Player File Edit View Window Help
100% Thu Jun 20 6:10:23 PM
*research_project - QGIS
Browser
Favorites
Home
/
/Volumes
GeoPackage
SpatiaLite
PostGIS
MSSQL
DB2
WMS/WMTS
XYZ Tiles
WCS
WFS
OWS
ArcGisMapServer
ArcGisFeatureServer
GeoNode
Layers
study_area
OpenStreetMap
Type to locate (⌘K)
1 legend entries removed.
Coordinate -97.6840,30.1717
Scale 1:81591
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326

SAT-API Browser

BUILD QUERY | RESULTS | CANDIDATES

Date Range

Start Date (required): 2019/07/14 06:33 PM

End Date (required): 2019/07/15 06:33 PM

Area Of Interest (required)

DRAW BBOX ON MAP

Property Filters

Add new filter

Select a property to filter

SUBMIT QUERY



```
$ sat-search search --intersects maine.geojson --found
1674 items found
$ clear
```

Cloud-friendly Data Formats

"Hey, let's not download the whole file everytime" - Eugene Cheipesh

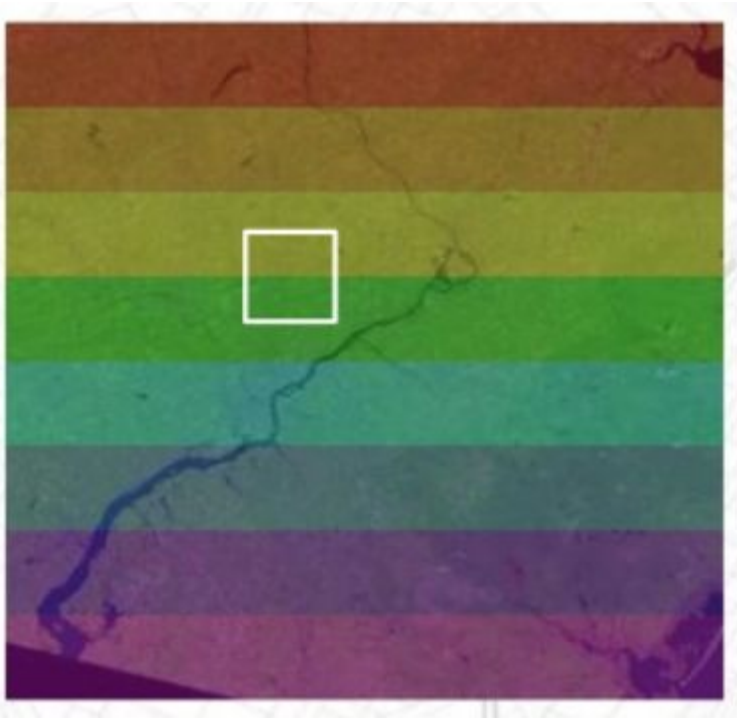
- Cloud-optimized GeoTIFF (COG)
- Meta Raster Format (MRF)
- Zarr
- Entwine Point Tiles
- TileDB

Cloud-Optimized GeoTIFF

- IFDs (Image File Directory) at beginning of file
  - reduces data seek times
- Uses Tiled Segments
  - decompress and load separately
- Overviews in file
  - efficiently use the data at any resolution

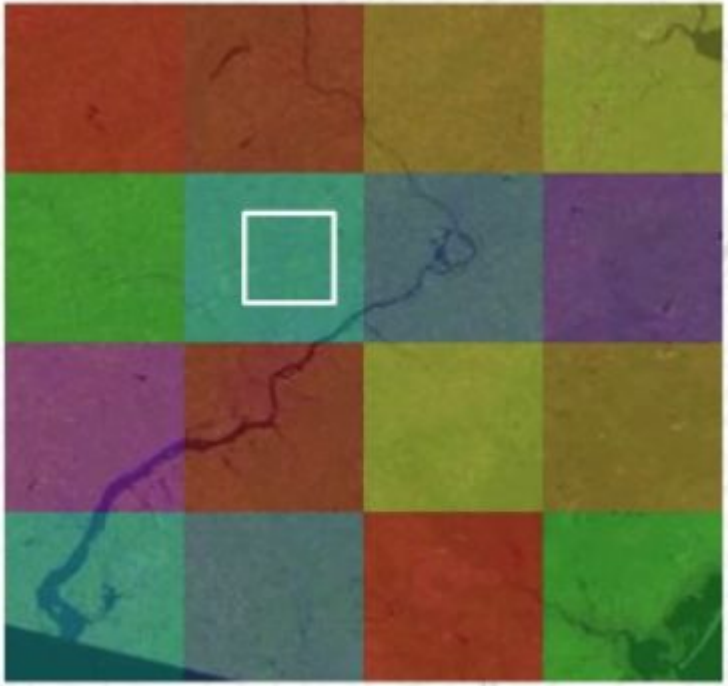
Slides and graphics courtesy of Eugene Cheipesh @azavea

TIFF Striped Segments = BAD



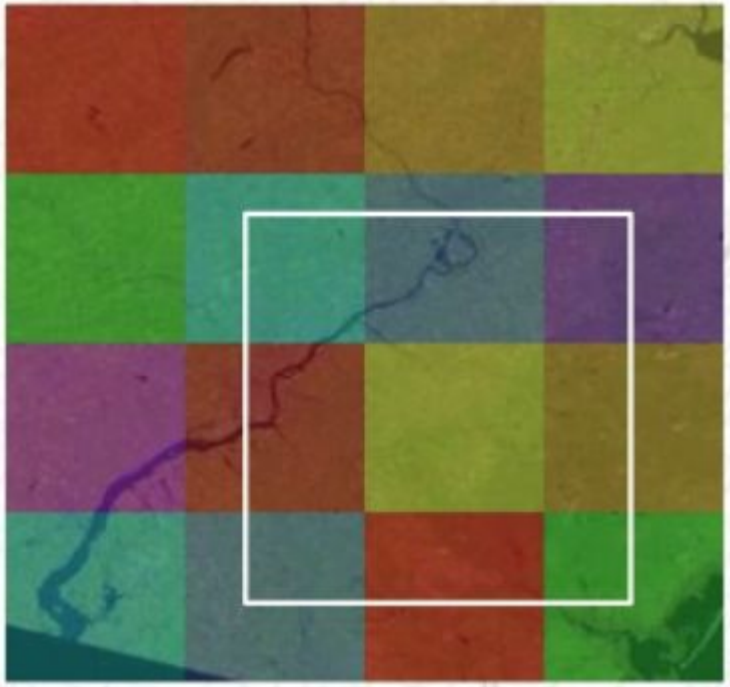


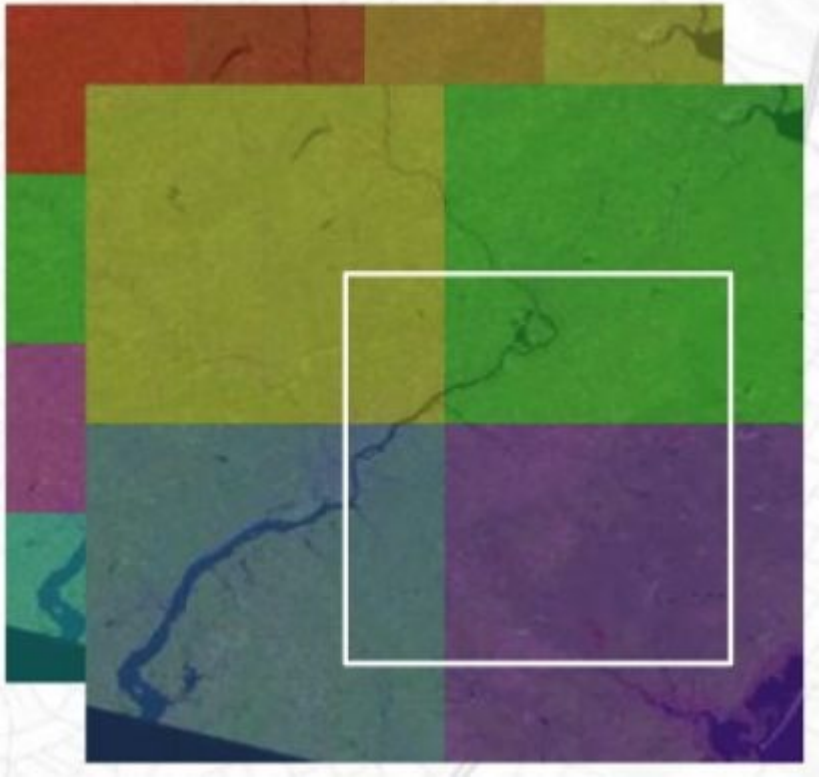
TIFF Tiled Segments = GOOD



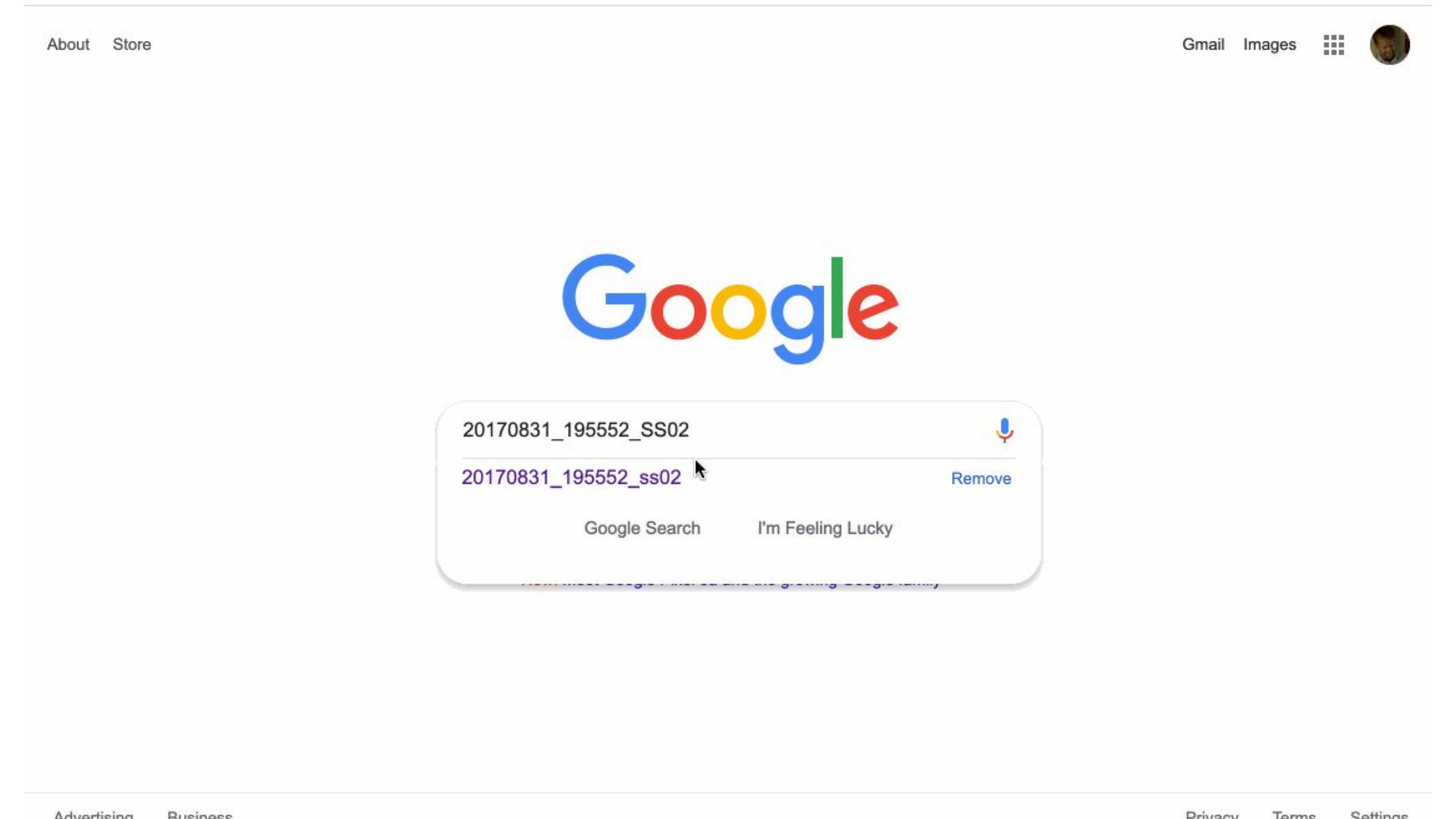


TIFF Tiled Segments = GOOD

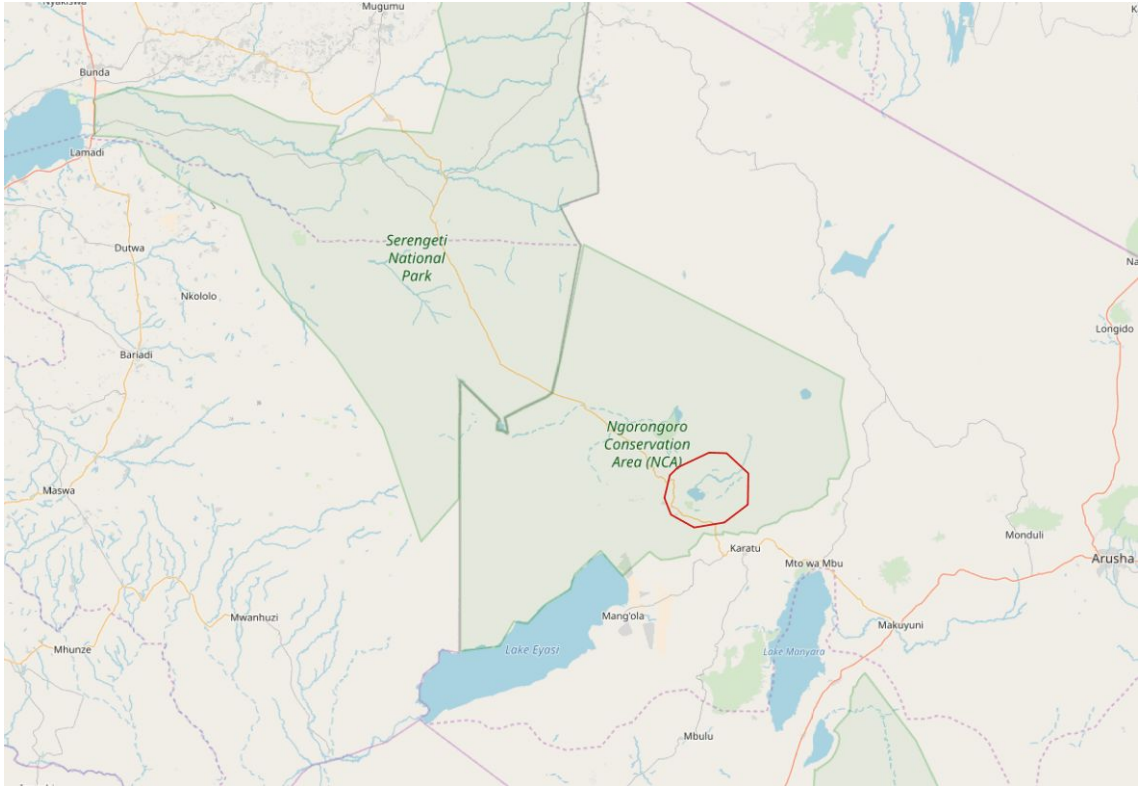




STAC
+
COG


About
Store
Gmail
Images
Google
20170831_195552_SS02
20170831_195552_ss02
Remove
Google Search
I'm Feeling Lucky
Advertising
Business
Privacy
Terms
Settings

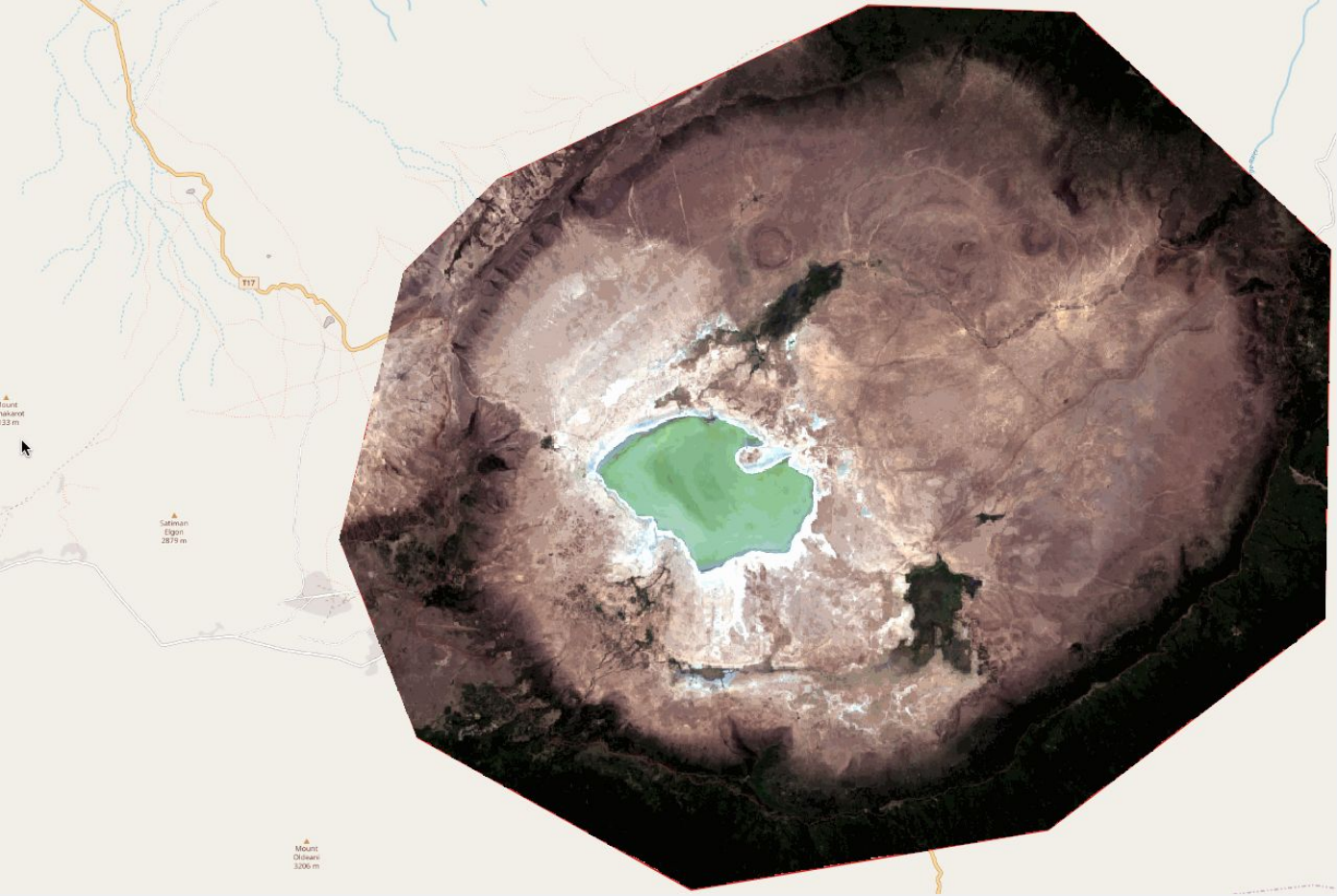
Ngorongoro crater: landsat-8 and sentinel-2



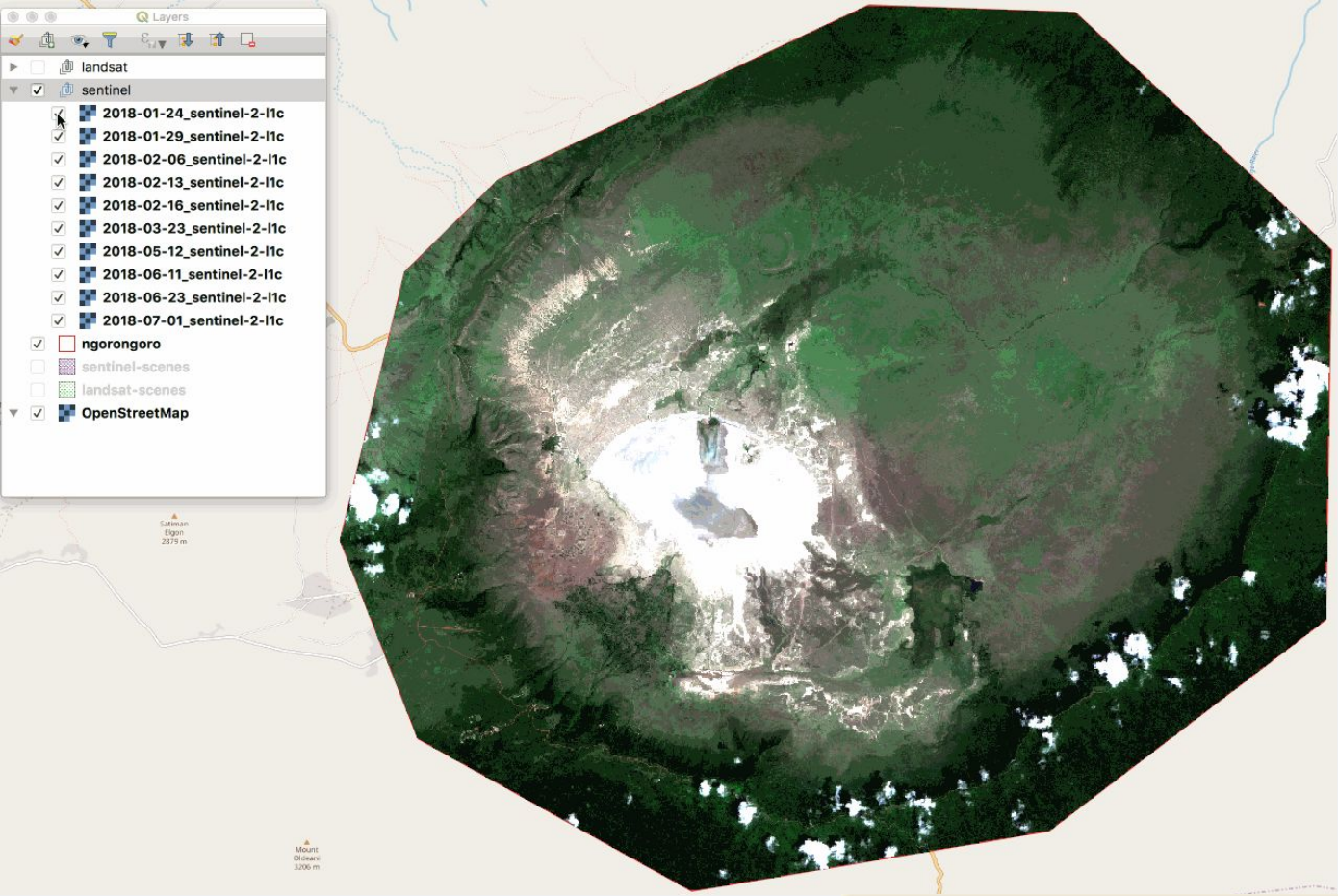
Ngorongoro crater: landsat-8 scenes

```
sat-search search
    \ --intersects ngorongoro.geojson --eo:cloud_cover 0/5 \
    -c landsat-8-l1 --save landsat-scenes.geojson \
    --print_md date eo:cloud_cover id
```

```
2014-12-11  5  LC08_L1TP_169062_20141211_20170416_01
2015-01-28  1  LC08_L1TP_169062_20150128_20170413_01
2015-02-13  0  LC08_L1TP_169062_20150213_20170413_01
2015-08-24  2  LC08_L1TP_169062_20150824_20170405_01
2015-09-09  1  LC08_L1TP_169062_20150909_20170404_01
2015-12-30  0  LC08_L1TP_169062_20151230_20170331_01
2016-02-16  0  LC08_L1TP_169062_20160216_20180526_01
2016-08-10  2  LC08_L1TP_169062_20160810_20170322_01
2016-10-13  0  LC08_L1TP_169062_20161013_20180526_01
2016-10-29  3  LC08_L1TP_169062_20161029_20170319_01
2017-01-17  0  LC08_L1TP_169062_20170117_20170311_01
2017-11-17  2  LC08_L1TP_169062_20171117_20171122_01
2018-02-05  0  LC08_L1TP_169062_20180205_20180221_01
2018-05-28  3  LC08_L1TP_169062_20180528_20180605_01
18 scenes found
```




```
$ time sat-fetch load landsat-scenes.geojson \
    --download red green blue
real 0m59.674s
```



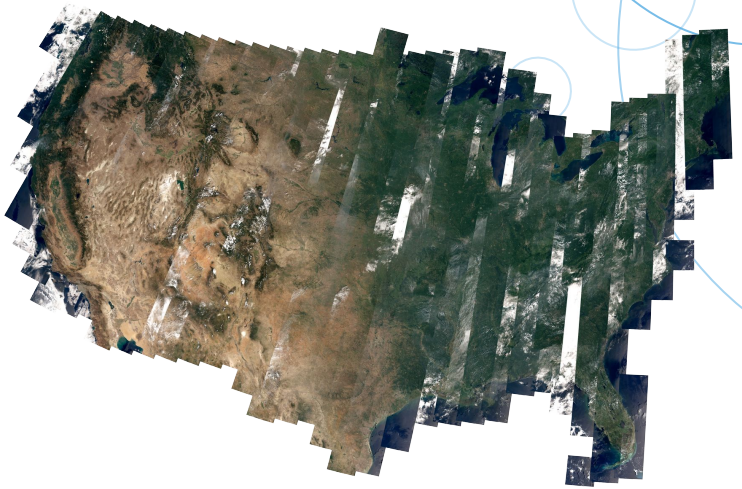

```
$ time sat-fetch load sentinel-scenes.geojson \
    --download red green blue
real 8m6.525s
```

E84 Earth Search

https://earth-search.aws.element84.com/stac

Public Search and Discovery API for Earth on AWS Datasets

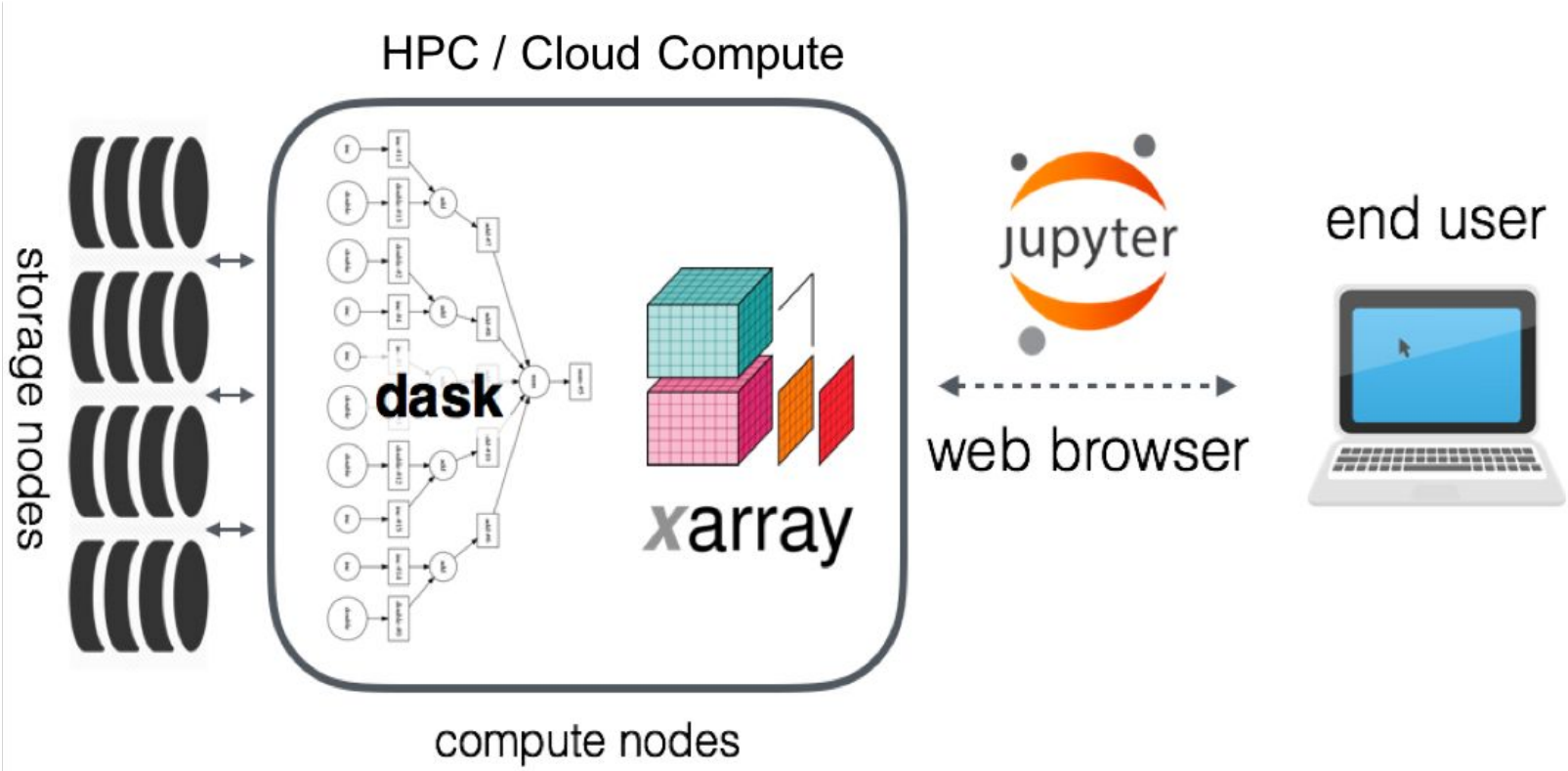
- Central search catalog and standardized metadata
- Single API that allows searching of STAC metadata for all Earth on AWS datasets
- Powered by sat-api (https://github.com/sat-utils/sat-api)
- Backed by an ElasticSearch index
- Proposed initial datasets
  - Sentinel-1
  - Sentinel-2
  - CBERS-4
  - Landsat-8
  - NAIP

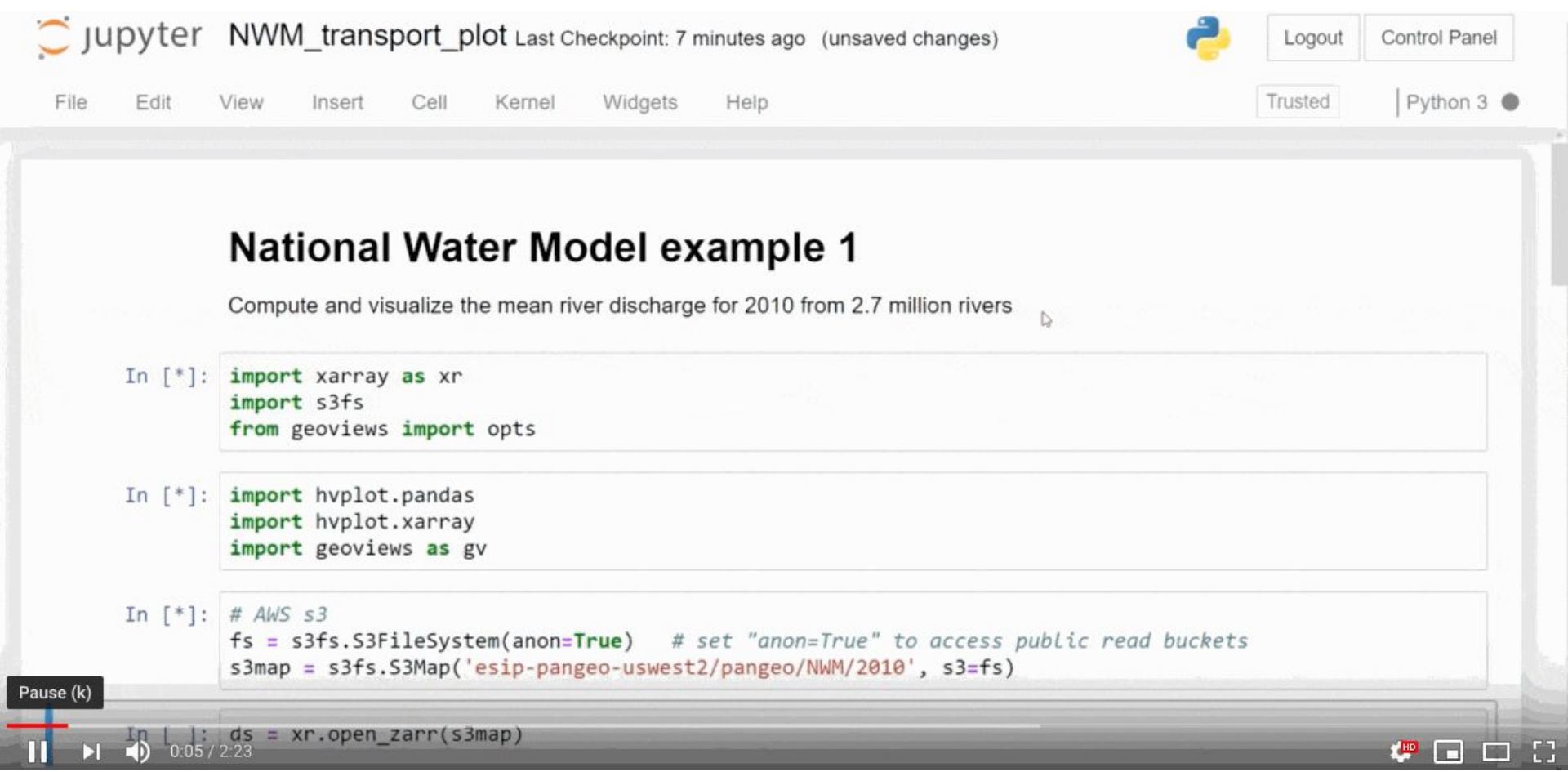


Landsat-8

Landsat is perhaps one of the most oft-cited and used EO datasets because of its historical archive (Landsat-8 was launched in 2013, but the entire Landsat archive goes back to 1982, with some caveats).

Pangeo


HPC / Cloud Compute
storage nodes
dask
xarray
compute nodes
jupyter
web browser
end user



courtesy of Rich Signell

This is possible because of

- Open Data in Cloud-friendly formats
- Open Software
- Open Standards
- Scientists doing Open Science

For updates, releases, news:

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Element 84

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